

HANDMATIC: WIRELESS ROBOTIC ARM CONTROL THROUGH GESTURE RECONGNITION

A PROJECT REPORT

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DECLARATION**

We affirm that the project report titled **“HANDMATIC: WIRELESS ROBOTIC ARM CONTROL THROUGH GESTURE RECONGNITION”** Submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering is the original work carried out by us. It has not formed part of any other project report or dissertation based on which a degree or award was conferred on an earlier occasion on this or any other candidates.

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ABSTRACT

This project focuses on designing and implementing an Arduino-based robotic arm for handling hazardous chemicals and performing dangerous tasks in environments where human presence is unsafe. The robotic arm operates wirelessly using RF (Radio Frequency) communication, allowing remote control from a safe distance. By eliminating human intervention in high-risk areas such as chemical processing plants, nuclear facilities, or toxic material handling sites, the system enhances safety and efficiency.

The robotic arm integrates sensor-based feedback to ensure precise operation. A gyroscope enables accurate orientation control, while a flex sensor detects detailed finger movements. These sensors allow the arm to mimic human dexterity, making it capable of performing delicate manipulations crucial for handling hazardous substances safely. The real-time sensor inputs ensure stable and accurate movements, reducing the risk of accidents.

The wireless RF module provides seamless communication between the transmitter and receiver, ensuring flexible and responsive control of the robotic arm. This setup enables users to operate the arm from a remote location without delays or signal interference, which is critical in hazardous environments.

To enhance security and access control, the system incorporates fingerprint authentication and a keypad-based multi-factor authentication system. These measures ensure that only authorized personnel can operate the robotic arm, preventing unauthorized access in sensitive environments.

By integrating robotics, wireless communication, advanced sensors, and security features, this project provides a safe, efficient, and reliable solution for executing hazardous industrial tasks remotely. The robotic arm significantly improves safety, precision, and operational efficiency, making it an ideal technology for chemical plants, hazardous material handling, and industrial automation where human presence poses substantial risks.

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LIST OF ABBREVIATIONS

LCD	Liquid Crystal Display
IC	Integrated Circuit
RF	Radio Frequency
RFID	Radio Frequency Identification
VR	Virtual Reality
ICSP	In-Circuit Serial Programming
AREF	Analog Reference
SRAM	Static Random Access Memory
EEPROM	Electrically Erasable Programmable Read-Only Memory
TTL	Time to Live
PWM	Pulse Width Modulation
VEE	Voltage at the Emitter
LED	Light Emitting Diode
UART	Universal Asynchronous Receiver Transmitter
GMD	General Material Designation
FLEX	Flexible Exchange Option

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The HandMatic project is a wireless robotic arm control system designed for hazardous environments such as chemical processing plants, nuclear facilities, and toxic material handling sites. It utilizes gesture-based control to enhance safety, precision, and efficiency in industrial automation. The system is built around an Arduino-controlled robotic arm, which operates wirelessly via RF communication, ensuring real-time response and reliable connectivity without the need for internet access.

The wearable transmitter device is the core of the system, equipped with a gyroscope sensor to track hand orientation and a flex sensor to detect finger movements. These sensors allow operators to control the robotic arm intuitively using natural hand gestures, enabling accurate manipulation of objects in hazardous environments. This eliminates the need for direct human intervention, reducing exposure to toxic chemicals, high temperatures, or radiation hazards.

To prevent unauthorized access, the system integrates fingerprint authentication and a keypad-based multi-factor security system. These security measures ensure that only authorized personnel can operate the robotic arm, which is crucial in sensitive industrial applications where precision and restricted access are essential.

The use of RF communication ensures low-latency, interference-resistant transmission between the user and the robotic arm. Unlike Wi-Fi-based systems, RF technology provides a dedicated, stable connection, making it ideal for industrial environments.

By combining gesture recognition, secure authentication, and wireless control, HandMatic offers a safer and more efficient alternative to manual operations in high-risk industries. The system significantly reduces occupational hazards while improving precision, operational efficiency, and workplace safety, making it a reliable solution for hazardous task automation.

1.2 OBJECTIVES OF THIS PROJECT

- **Enhance Workplace Safety** – Reduce human exposure to hazardous environments such as chemical processing plants, nuclear facilities, and toxic material handling sites by enabling remote operation of a robotic arm.
- **Develop a Gesture-Based Control System** – Implement an intuitive and natural control mechanism using a gyroscope and flex sensor to translate hand movements into robotic arm actions for precise and effortless operation.
- **Ensure Secure Access Control** – Integrate fingerprint authentication and keypad-based multi-factor authentication to restrict access and ensure that only authorized personnel can operate the robotic arm.
- **Enable Reliable Wireless Communication** – Utilize RF communication for low-latency, interference-resistant data transmission, ensuring real-time and uninterrupted control of the robotic arm without relying on Wi-Fi or internet connectivity.
- **Improve Industrial Efficiency** – Enhance precision and operational control in hazardous industries, allowing for accurate manipulation of objects while minimizing the risks associated with manual handling.
- **Reduce Occupational Hazards** – Provide a safe alternative to traditional manual operations, significantly lowering the chances of chemical exposure, radiation contamination, and other workplace injuries.

1.3 NEED OF THE PROJECT

The HandMatic project enhances worker safety, precision, and efficiency in hazardous industrial environments such as chemical plants, nuclear facilities, and toxic material handling sites. These industries expose workers to dangerous substances, extreme temperatures, and radiation, posing serious health risks. A remotely controlled robotic arm minimizes human intervention, reducing injuries, contamination, and workplace accidents. Manual handling of hazardous materials increases the risk of spills, explosions, and chemical reactions due to human error. By using gesture-controlled automation, HandMatic enables precise object manipulation, improving accuracy and operational safety.

Another key necessity of this project is remote machinery operation. In unsafe environments, a wireless RF-based robotic arm establishes a real-time, interference-resistant communication link, allowing workers to control tasks from a secure location. Security is also critical in hazardous industries, where unauthorized access to sensitive equipment can lead to accidents. To prevent this, HandMatic integrates fingerprint authentication and keypad-based multi-factor security, ensuring only authorized personnel can operate the system.

With the rise of Industry 4.0 and automation, there is growing demand for cost-effective robotic solutions. Many industrial robots are expensive and complex, limiting accessibility for smaller industries. HandMatic offers a low-cost, Arduino-based alternative, making advanced automation affordable and scalable. By integrating gesture recognition, wireless communication, and secure authentication, the project provides an innovative, reliable, and efficient solution for hazardous task automation. Ultimately, HandMatic enhances workplace safety, boosts productivity, and improves industrial efficiency.

1.4 SCOPE OF THE PROJECT

The HandMatic project is designed to enhance automation and safety in hazardous industrial environments by providing a secure, intuitive, and efficient robotic arm control system. It primarily targets industries where manual operations pose significant health risks, such as chemical processing plants, nuclear facilities, and hazardous material handling sites. By enabling gesture-based, wireless control, the project reduces direct human exposure to dangerous substances, extreme temperatures, and radiation.

The scope of the project includes:

- **Industrial Applications** – The system is beneficial in chemical, pharmaceutical, nuclear, mining, and defense sectors, where remote handling of materials is necessary.
- **Security and Access Control** – Fingerprint authentication and multi-factor security ensure restricted access to the robotic arm, preventing unauthorized operation.
- **Wireless Communication** – The RF-based control mechanism ensures low-latency and interference-resistant operation without relying on internet connectivity.
- **Cost-Effective Automation** – The system provides an affordable, Arduino-based alternative to expensive industrial robots, making automation more accessible.
- **Scalability and Adaptability** – The modular design allows for future enhancements, such as AI-based gesture recognition, extended wireless range, and improved robotic arm mechanics.

1.5 ADVANTAGES OF THE PROJECT

The HandMatic system offers several key advantages over traditional robotic control methods:

1. Enhanced Workplace Safety

- Reduces human exposure to toxic chemicals, extreme heat, and radiation.
- Lowers the risk of accidents, contamination, and workplace injuries.

2. Intuitive Gesture-Based Control

- Uses gyroscope and flex sensors to translate hand movements into robotic arm actions.
- Provides a natural and user-friendly method of control without complex programming.

3. Secure Access and Authentication

- Integrates fingerprint authentication and multi-factor security to prevent unauthorized use.
- Ensures that only trained personnel can operate the robotic system.

4. Reliable Wireless Communication

- RF technology provides a dedicated, stable connection without requiring internet access.
- Low latency ensures real-time control and precise operation.

5. Cost-Effective and Scalable

- Uses Arduino and off-the-shelf components, making it affordable for small and medium-scale industries.
- Can be customized and expanded to support additional sensors, actuators, and communication protocols.

6. Increased Productivity and Efficiency

- Reduces manual labor in hazardous tasks, increasing overall operational efficiency.
- Minimizes material waste and human errors, ensuring accurate handling of objects.

1.6 ORGANIZATION OF THE REPORT

Chapter 1 provides the overview, objective, scope and advantages of the project.

Chapter 2 presents the details of the various literatures reviewed in relevant to the proposed work.

Chapter 3 provides details of Gesture- Controlled wireless Robotic Arm in the proposed system.

Chapter 4 provides a detailed discussion of the result obtained through the laboratory prototype model.

Chapter 5 Finally, the conclusion and future scope of the project work is discussed.

1.7 SUMMARY

The HandMatic project introduces a secure, intuitive, and efficient wireless robotic arm control system designed for hazardous industrial environments. By integrating gesture-based control, secure authentication, and real-time RF communication, the system significantly improves workplace safety, operational efficiency, and automation capabilities.

This chapter outlined the project's objectives, scope and advantages. With advancements in AI, wireless connectivity, and industrial automation, HandMatic has the potential to evolve into a cutting-edge robotic solution for various industries. Future improvements will focus on enhancing security, expanding connectivity options, and integrating AI-driven control mechanisms to further refine the system's performance and reliability.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

The advancement of robotic arms for hazardous environments has been extensively researched, focusing on gesture-based control, wireless communication, and security. Earlier robotic arms relied on joysticks or wired systems, limiting flexibility and efficiency. With the rise of wireless communication, technologies like Wi-Fi, Bluetooth, and RF modules have been explored. While Wi-Fi-based systems offer long-range communication, they are prone to network interference and security risks. In contrast, RF-based systems provide low-latency and interference-resistant communication, making them more reliable for real-time industrial applications.

Studies on gesture recognition highlight the effectiveness of gyroscopes and accelerometers in detecting hand movements for intuitive robotic control. Additionally, flex sensors have been successfully implemented to recognize finger bending movements, allowing precise robotic manipulation. This enhances usability, making automation more natural and efficient.

Security is a major concern in industrial robotics. While password-based authentication has been used, biometric security like fingerprint authentication offers greater protection. Multi-factor authentication, combining fingerprint scanning and keypad input, further strengthens security, preventing unauthorized access. Building on these findings, the HandMatic project integrates gesture control, RF-based communication, and multi-factor authentication, providing a secure, efficient, and reliable robotic solution for hazardous industrial applications.

2.2 LITERATURE REVIEW

Singh et al. (2024) have developed a wireless robotic arm control system using a hand glove controller for material handling in industrial applications. The study emphasizes the increasing trend of robotic integration in industries to reduce human labor and improve efficiency. The proposed system utilizes an accelerometer as an inertial sensor to detect hand gestures, which are then processed by a microcontroller in the transmitter unit. The sensor data is wirelessly transmitted using an RF transceiver module to the robotic arm's receiver, enabling gesture-based movement control. This system enhances object grasping and positioning capabilities, making it useful for industrial and medical applications.

Wang et al. (2024) have developed a gesture recognition system using MEMS accelerometers to enable human-computer interaction and robotic control. The system integrates an ARM processor, ADXL345 accelerometer, NRF24L01 wireless module, and LABVIEW for data processing and visualization. Five accelerometers placed on the fingers collect acceleration data, which is filtered and compared against predefined thresholds to determine finger positions. The processed data is transmitted wirelessly to a receiver, which identifies gestures and sends the results to LABVIEW via a serial port. This system allows users to control web pages, image displays, and robotic grippers, highlighting its potential for gesture-based automation and interactive control.

Anughna et al. (2020) have developed a wireless robotic arm model that integrates flex and gyro sensors for precise and real-time control. Flex sensors detect finger movements, while gyro sensors measure the orientation and motion of the arm. This data is wirelessly transmitted, eliminating the need for physical connections and enhancing mobility and flexibility in remote operations. The study highlights its applications in medical surgery, hazardous material handling, and industrial automation, where accurate and responsive control is crucial. The paper discusses the system's components, working principles, and advantages, emphasizing the benefits of wireless robotic technology in modern automation.

Baviskar et al. (2019) have proposed a fingerprint-based security system for remote-controlled robotics, ensuring secure access and operation. The study addresses the risks of misuse, theft, and unauthorized control in robotics and IoT systems by integrating biometric authentication. The designed system responds only to commands from authenticated users, enhancing security in applications ranging from domestic automation to military operations. The paper highlights the importance of secure data transmission algorithms in preventing cyber threats. This research contributes to the development of personalized and secure future robotics, ensuring controlled access and reliable remote operations.

Dheeban et al. (2018) developed an Arduino-controlled gesture robot that converts hand gestures into robotic movements using sensors. The system detects gestures, processes them via an Arduino micro controller, and controls the robot's actions. The study explores hardware components, sensor integration, and software algorithms for gesture recognition. With low-cost implementation, the research highlights applications in assistive devices, education, and interactive robotics, demonstrating the potential for gesture-based human-robot interaction.

Komol et al. (2018) have developed a dual security system integrating RFID and fingerprint recognition for secure access control in door lock operations. By combining RFID technology with biometric authentication, the system ensures robust security, allowing entry only when both an RFID card and a fingerprint scan are verified. This two-step authentication prevents unauthorized access and enhances reliability. The study details the architecture, design, and embedded system integration of the security mechanism. The proposed system is applicable in residential, commercial, and industrial settings, offering a secure and efficient access control solution.

2.3 SUMMARY

This chapter explores advancements in gesture-based robotic arm control, wireless communication, and security for industrial applications. RF-based communication is highlighted as a reliable alternative to Wi-Fi due to its low latency and interference resistance. Gesture recognition using accelerometers, gyroscopes, and flex sensors has been widely studied for intuitive robotic control. Security concerns in robotics are addressed through fingerprint authentication and multi-factor authentication to prevent unauthorized access. Various studies on wireless robotic arms, MEMS-based gesture systems, and secure access mechanisms are reviewed. Building on these findings, the HandMatic project integrates gesture control, RF communication, and multi-factor authentication to enhance safety and efficiency in hazardous environments.

CHAPTER 3

HANDMATIC: GESTURE-CONTROLLED WIRELESS ROBOTIC ARM

3.1 INTRODUCTION

The HandMatic: Gesture-Controlled Wireless Robotic Arm is designed to perform tasks in hazardous environments where human presence is risky. Using wireless RF communication, it allows remote operation, ensuring safety in industries like chemical processing and toxic material handling. The system integrates gyroscopes and flexes sensors for precise movement, replicating human-like dexterity. Enhanced with fingerprint authentication and keypad security, HandMatic ensures only authorized users control it, making it a secure and efficient solution for industrial automation.

3.2 EXISTING SYSTEM OF WIRELESS ROBOTIC ARM

This project focuses on developing a versatile, gesture-controlled all-terrain surveillance robot equipped with a robotic arm and First-Person View (FPV) cameras for remote monitoring. Unlike traditional operation-specific robots, this system integrates multiple functions, making it suitable for surveillance in remote or hazardous environments where human intervention is difficult. The robot's locomotion and robotic arm movements are controlled using hand gestures, enhancing ease of operation.

The FPV system, featuring a VR-compatible setup with dual cameras on a pan-and-tilt mechanism, provides a panoramic view for effective monitoring. Designed with off-road tires and a conveyor belt, the robot ensures smooth movement across various terrains. This system is valuable for industrial workstations, research applications, and data collection in challenging environments.

In figure 3.1 shows the block diagram illustrates how commands are transmitted for smooth robot functioning. Gesture-controlled transmitters and receivers manage locomotion and the robotic arm. A glove-based controller mimics hand movements for intuitive control. FPV cameras provide high-resolution, real-time surveillance, integrated with a pan-tilt mechanism and servo motor for head- tracking via a VR set. The robotic arm mirrors human hand joints, making object manipulation easy. The compact FPV camera with a built-in transmitter ensures seamless connectivity and efficient monitoring.

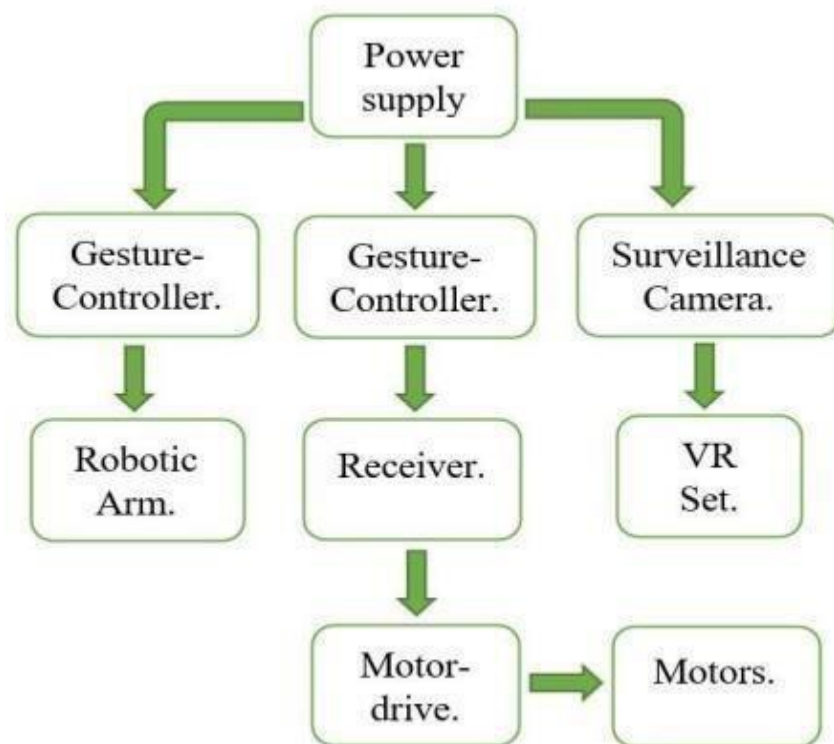


Figure 3.1 Block Diagram for Existing System

3.2.1 Limitations of Existing System

1. Gesture Recognition Accuracy:

- The system's reliance on hand gestures for control may face issues related to **gesture recognition accuracy**. Hand gestures can be interpreted differently based on various factors, such as lighting conditions, the angle of the hand, or the user's unique hand movements, which may lead to unintended actions or system errors.

- The abstract doesn't mention whether the system uses advanced gesture recognition algorithms or if it has the ability to handle errors, such as false positives or negatives in the movement commands.

2. User Fatigue:

- **User fatigue** may become a concern during prolonged use, especially when controlling the robotic arm or locomotion over extended periods. Constant hand movement can strain the user's muscles, leading to discomfort or reduced effectiveness in controlling the robot.

3. Limited Range of Gesture Inputs:

- The abstract mentions controlling the robotic arm by moving fingers and wrist, but it does not clarify the **range of gestures** supported. If the system is limited to a small set of predefined gestures, the user may find it cumbersome to perform more complex tasks.
- It's also unclear whether the system supports multi-finger or multi-joint gestures for more nuanced control or if it only interprets basic movements.

4. Complexity in Training:

- The system's design implies that users must familiarize themselves with specific hand and wrist motions. While the design simplifies control, the **learning curve** could still be an issue, especially if users need to memorize a set of gestures or movements for precise control of the robotic arm and FPV camera.
- It's unclear whether there is any form of training mode or feedback for the user to learn these gestures effectively.

5. Limited to Simple Tasks:

- The abstract focuses on basic locomotion and arm control, but it doesn't address the **complexity of tasks** the robotic arm can handle. If the robotic arm is only capable of simple movements like grabbing or lifting, it may not be versatile enough for more intricate tasks or industrial applications.
- It's not clear whether the system supports advanced features such as fine motor control or object manipulation with high precision.

3.3 PROPOSED HANDMATIC: GESTURE-CONTROLLED WIRELESS ROBOTIC ARM

The Gesture-Controlled Wireless Robotic Arm is designed to enhance safety and efficiency in hazardous environments by enabling remote operation through gesture-based control. It is particularly useful in industries dealing with chemical processing, toxic material handling, and industrial automation, where human presence poses significant risks.

This project consists of two main sections: the transmitter and the receiver sides. The transmitter side includes gyroscopes and flexes sensors embedded in a wearable device, capturing hand movements and translating them into commands. These commands are then transmitted wirelessly via RF communication to the receiver side, where the robotic arm interprets the signals and performs corresponding actions with precision. This ensures human-like dexterity, improving accuracy and responsiveness in delicate and complex tasks.

To enhance security, HandMatic incorporates fingerprint authentication and a keypad-based multi-factor security system at the transmitter side, ensuring that only authorized personnel can operate the system. By combining wireless communication, real-time gesture recognition, and robust security features, HandMatic offers a secure, efficient, and user-friendly solution for industries requiring reliable robotic automation.

3.3.1 Proposed Transmitter System

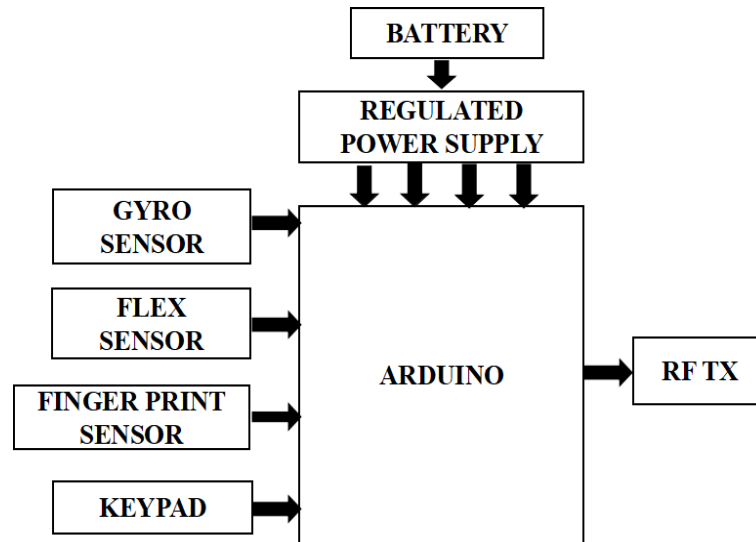


Figure 3.2 Block Diagram for Proposed Transmitter System

In Figure 3.2 shows our system, the Arduino acts as the central controller, processing inputs from various sensors and devices to control the motor's operation via RF communication. The inputs include data from a gyro sensor (for detecting orientation or movement), a fingerprint sensor (for authentication and security), and a keypad (for user commands). The Arduino then sends processed commands through an RF transmitter to a remote receiver, which could be another Arduino or a dedicated controller. This RF communication allows the system to operate wirelessly, providing flexibility in the control of the motor and ensuring that the motor's actions depend on valid sensor readings or user inputs. Power for the entire system is provided by a battery, ensuring portability and standalone operation.

3.3.2 Proposed Receiver System

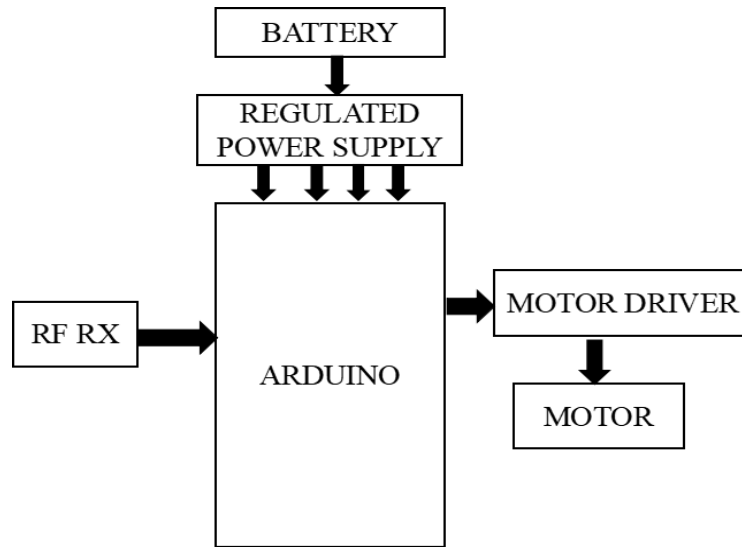


Figure 3.3 Block Diagram for Proposed Receiver System

In Figure 3.3 shows the receiving end, the RF receiver collects the transmitted data and forwards it to a motor driver, which then controls the motor based on the received commands. The motor driver interprets signals such as start/stop, speed, and direction, allowing the motor to perform the desired actions. The system can be used in various applications like robotic movement, remote controlled devices, or security systems, where precise control and authentication are crucial. A battery- powered setup ensures the system remains portable and independent of fixed power sources, making it ideal for mobile or outdoor use.

3.4 GYRO SENSOR

The gyroscope sensor, as shown in Figure 3.4, plays a crucial role in the Gesture-Controlled Wireless Robotic Arm by enabling precise motion tracking and orientation detection. It measures angular velocity and detects changes in hand movements, allowing the robotic arm to accurately mimic the operator's gestures. Integrated into the transmitter side, the gyroscope captures real-time motion data and sends it via RF communication to the receiver side, where the robotic arm interprets and executes the corresponding movements. This ensures smooth, responsive, and intuitive control, making the system highly effective for delicate and hazardous tasks.

The gyroscope's high sensitivity and accuracy enhance the precision of the robotic arm, enabling it to handle complex industrial operations with human-like dexterity. By integrating this advanced sensor, the HandMatic system significantly improves the efficiency and safety of remote-controlled automation in hazardous environments like chemical processing plants and industrial facilities.

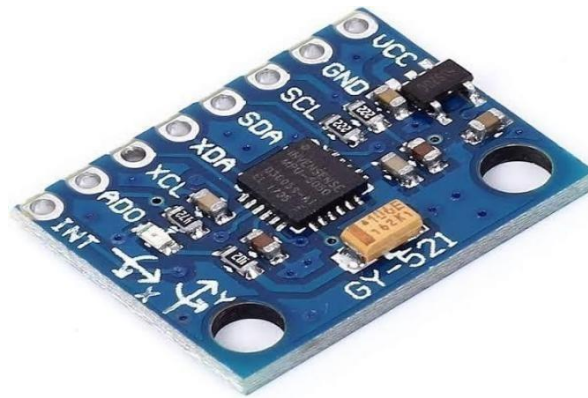


Figure 3.4 Gyro Sensor

3.5 FLEX SENSOR

A flex sensor is a specialized sensor used to measure the amount of bending or deflection, making it a crucial component in various applications. It is commonly found in robotics, wearable technology, door sensors, and interactive toys like Nintendo's Power Glove. This sensor consists of a plastic strip coated with a carbon surface, where the resistance changes when the strip bends. The resistance variation is directly proportional to the degree of bending, allowing it to function as a goniometer for precise angle measurement. In the Gesture- Controlled Wireless Robotic Arm, the flex sensor is integrated into the transmitter side, where it detects finger movements and transmits data via RF communication to the receiver side. The robotic arm then interprets these signals and replicates the hand gestures with high accuracy and responsiveness.

This ensures a smooth and intuitive control mechanism, making the system effective for applications in hazardous environments, medical fields, and industrial automation. A flex sensor Figure 3.5 measures bending or deflection by changing its resistance when bent. It consists of a plastic strip with a carbon surface, where resistance varies proportionally to the bending angle. In the HandMatic system, flex sensors capture finger movements, enabling precise gesture-based control of the robotic arm.



Figure 3.5 Flex Sensor

3.5.1 Types of Flex Sensor

Flex sensors are primarily classified into two types based on their size:

- **2.2-inch Flex Sensor** – This compact sensor is commonly used in applications requiring precise resistance control, such as computer interfaces, rehabilitation devices, servo motor control, security systems, and music interfaces.
- **4.5-inch Flex Sensor** – This larger sensor offers higher resistance variation and is used in applications demanding greater flexibility and sensitivity, including robotic systems, automation projects, and wearable technology.

Both types function on the principle that resistance increases as the sensor bends, making them effective for detecting movement and deflection. (Refer to Figure 3.6).

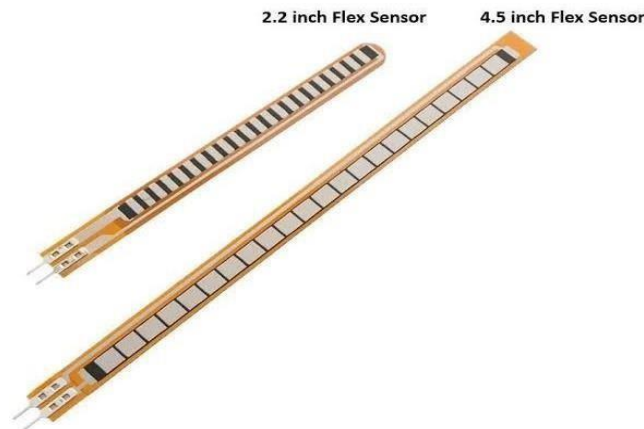


Figure 3.6 2.2-inch and 4.5 inch Flex Sensor

3.5.2 Pin Configuration

The pin configuration of the flex sensor is shown below. It is a two-terminal device, and the terminals are like p1 & p2. This sensor doesn't contain any polarized terminal such as diode otherwise capacitor, which means there is no positive & negative terminal. The required voltage of this sensor to activate the sensor ranges from 3.3V -5V DC which can be gained from any type of interfacing.

- Pin P1: This pin is generally connected to the +ve terminal of the power source.
- Pin P2: This pin is generally connected to GND pin of the power source.

3.5.3 Specifications and Features

The specifications and features of this sensor include the following.

- Operating voltage of this sensor ranges from 0V to 5V.
- It can function on low-voltages.
- Power rating is 1 Watt for peak & 0.5Watt for continuous.
- Operating temperature ranges from -45°C to +80°C.
- Flat resistance is 25K Ω .

- The tolerance of resistance will be $\pm 30\%$.
- The range of bend resistance will range from 45K -125K Ohms.

3.5.4 Applications

The applications of the flex-sensor include the following.

- Medical Instruments
- Peripherals of Computer
- Robotics
- Physical Therapy
- Virtual Motion (Gaming)
- Musical Instruments

Thus, this is all about flex sensor. From the above information, we can finally conclude that when this sensor is bent then the terminal resistance of this sensor will be changed.

3.6 FINGER PRINT SENSOR

A fingerprint sensor is a biometric device used to authenticate and verify an individual's identity based on unique fingerprint patterns. It captures a person's fingerprint image and converts it into a digital template for authentication.

The sensor works by scanning the ridges and valleys of a fingerprint using optical, capacitive, or ultrasonic technology. The collected fingerprint data is then compared with stored templates in the system's memory to grant or deny access. Fingerprint sensors are widely used in security systems, mobile devices, access control systems, and industrial automation to ensure authorized usage. These sensors provide high security, reliability, and quick authentication, making them ideal for applications where secure and restricted access is required. In robotics, fingerprint sensors can be integrated for multi-factor authentication, ensuring only authorized users can control sensitive systems. Figure 3.7 illustrates the fingerprint sensor in a security setup.

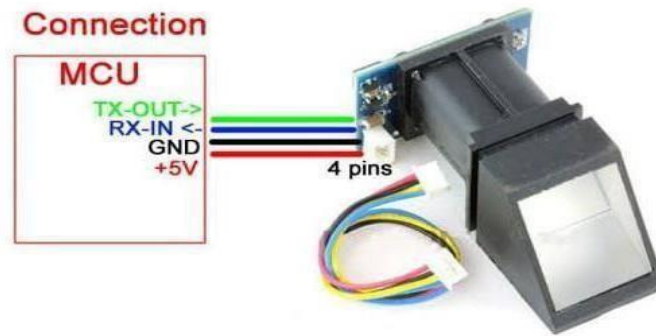


Figure 3.7 Finger Print Sensor

3.6.1 Working Principle of Fingerprint Sensor

A fingerprint sensor operates by capturing and analyzing the unique ridge patterns of a fingerprint to verify a user's identity. The working principle depends on the type of sensor used, such as optical, capacitive, or ultrasonic sensors.

- 1. Image Acquisition** – When a user places their finger on the sensor, it captures an image of the fingerprint using light (optical), electrical signals (capacitive), or sound waves (ultrasonic).
- 2. Feature Extraction** – The sensor processes the image to identify key fingerprint features, such as ridges, valleys, and minutiae points.
- 3. Data Conversion** – The extracted features are converted into a digital template using an algorithm.
- 4. Authentication & Matching** – The generated template is compared with stored fingerprint templates in the system's memory. If a match is found, access is granted; otherwise, it is denied.

This efficient, highly secure, and contact-based biometric technology is widely used for authentication in robotics, security systems, and industrial automation.

3.7 KEYPAD SENSOR

A keypad sensor is an input device used for entering numeric or alphanumeric data into a system. It consists of multiple keys arranged in a matrix format (typically 4×3 or 4×4) and is widely used in security systems, ATM s, access control, and embedded applications.

When a key is pressed, the keypad detects the electrical connection between the corresponding row and column, sending a unique signal to the micro controller. The microcontroller processes this signal and determines the pressed key based on a predefined lookup table. The keypad sensor is simple, reliable, and efficient, making it suitable for applications requiring PIN-based authentication, device control, and data entry.

In the HandMatic system, the keypad sensor enhances security by providing an additional layer of authentication alongside the fingerprint sensor, ensuring only authorized users can control the robotic arm. Figure 3.8 illustrates the keypad sensor used in this project.



Figure 3.8 4x4 Matrix Keypad Sensor

3.7.1 Features of Keypad Sensor

- **Multiple Key Input:** Supports 4×3 or 4×4 matrix configurations for efficient data entry.
- **Reliable Operation:** Uses mechanical or membrane switches for durability.
- **Low Power Consumption:** Suitable for battery-operated devices.
- **Easy Interface:** Works seamlessly with micro controllers like Arduino.
- **Enhanced Security:** Used in PIN-based authentication and access control systems.

3.7.2 Key Specification

Key specifications of a keypad sensor include matrix configuration (4×3 or 4×4), operating voltage (3.3V–5V), and low power consumption for efficient performance. It features durable membrane or mechanical keys, a fast response time, and compatibility with micro controllers like Arduino. The sensor is ideal for secure access control and automation applications.

- **Maximum Rating:** 24 VDC, 30 mA.
- **Interface:** 8-pin access to 4x4 matrix.
- **Operating temperature:** 32 to 122 °F (0 to 50°C).
- **Dimensions:** Keypad, 2.7 x 3.0 in (6.9 x 7.6 cm).
- **Cable:** 0.78 x 3.5 in (2.0 x 8.8 cm).

3.7.3 Applications

Keypad sensors are widely used in access control systems, ATM machines, security locks, industrial automation, and home automation. They are also employed in electronic voting machines, password-protected devices, smart safes, and IoT-based security systems. Their reliability and ease of integration make them ideal for various authentication and input-based applications.

- Security systems
- Menu selection
- Data entry for embedded systems

3.7.4 Working Principle of Keypad Sensor

The working principle of a keypad sensor is based on detecting key presses through electrical connections. It consists of a matrix of rows and columns, where each key is placed at the intersection of a row and a column. When a key is pressed, it creates a connection between a specific row and column, allowing current to flow.

A micro controller continuously scans the keypad by sending signals to the rows and detecting responses from the columns. Based on the signal received, the pressed key is identified, and the corresponding action is executed. Keypad sensors are widely used for secure input systems, authentication, and control applications.

Matrix keypads use a combination of four rows and four columns to provide button states to the host device, typically a micro controller. Underneath each key is a push-button, with one end connected to one row, and the other end connected to one column. These connections are shown in Figure 3.9.

In order for the micro controller to determine which button is pressed, it first needs to pull each of the four columns (pins 1-4) either low or high one at a time, and then poll the states of the four rows (pins 5-8). Depending on the states of the columns, the micro controller can tell which button is pressed.

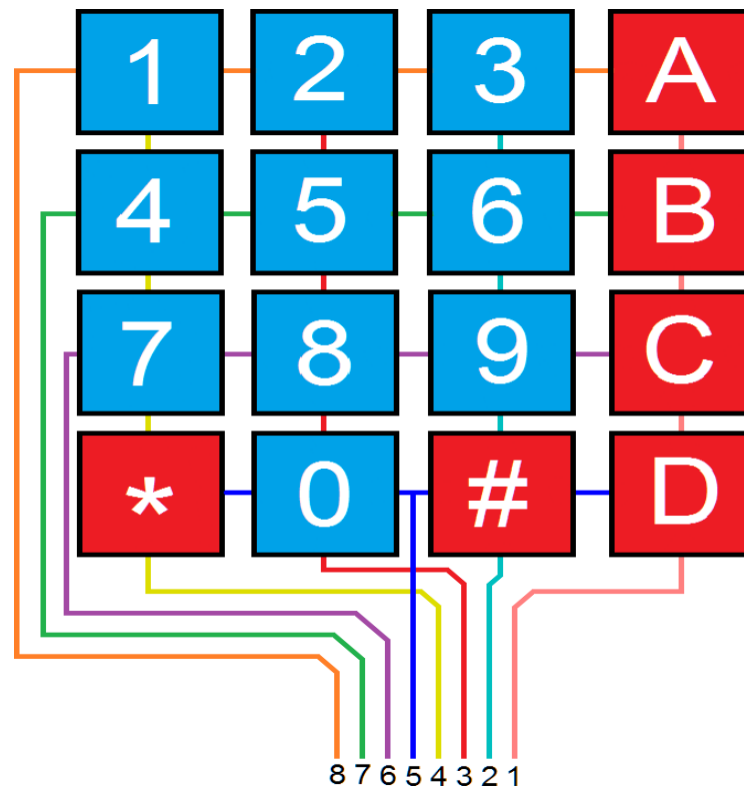


Figure 3.9 Matrix Keypad Connections

For example, say your program pulls all four columns low and then pulls the first row high. It then reads the input states of each column, and reads pin 1 high. This means that a contact has been made between column 4 and row 1, so button ‘A’ has been pressed.

3.8 AUDUINO UNO

The Arduino Uno is a micro controller board based on the ATmega328. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the micro controller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega8U2 programmed as a USB-to-serial converter. "Uno" means one in Italian and is named to mark the upcoming release of Arduino 1.0. The Uno and version 1.0 will be the reference versions of Arduino, moving forward. The Uno is the latest in a series of USB.

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The Uno and version 1.0 will be the reference versions of Arduino, moving forward. The Uno is the latest in a series of USB Arduino boards, and the reference model for the Arduino platform; for a comparison with previous versions.

The Arduino Uno, shown in Figure 3.10, is a widely used micro controller board based on the ATmega328P. It features 14 digital input/output pins, 6 analog inputs, a 16 MHz quartz crystal, a USB connection, a power jack, and a reset button. The board operates at 5V and can be powered via USB or an external power source. It is commonly used in robotics, automation, IoT, and embedded systems due to its ease of use, open-source nature, and extensive community support. The Arduino IDE simplifies programming, making it an excellent choice for both beginners and experts. The Arduino Uno supports various sensors, actuators, and communication modules, providing a flexible and reliable platform for developing interactive and real-time systems.

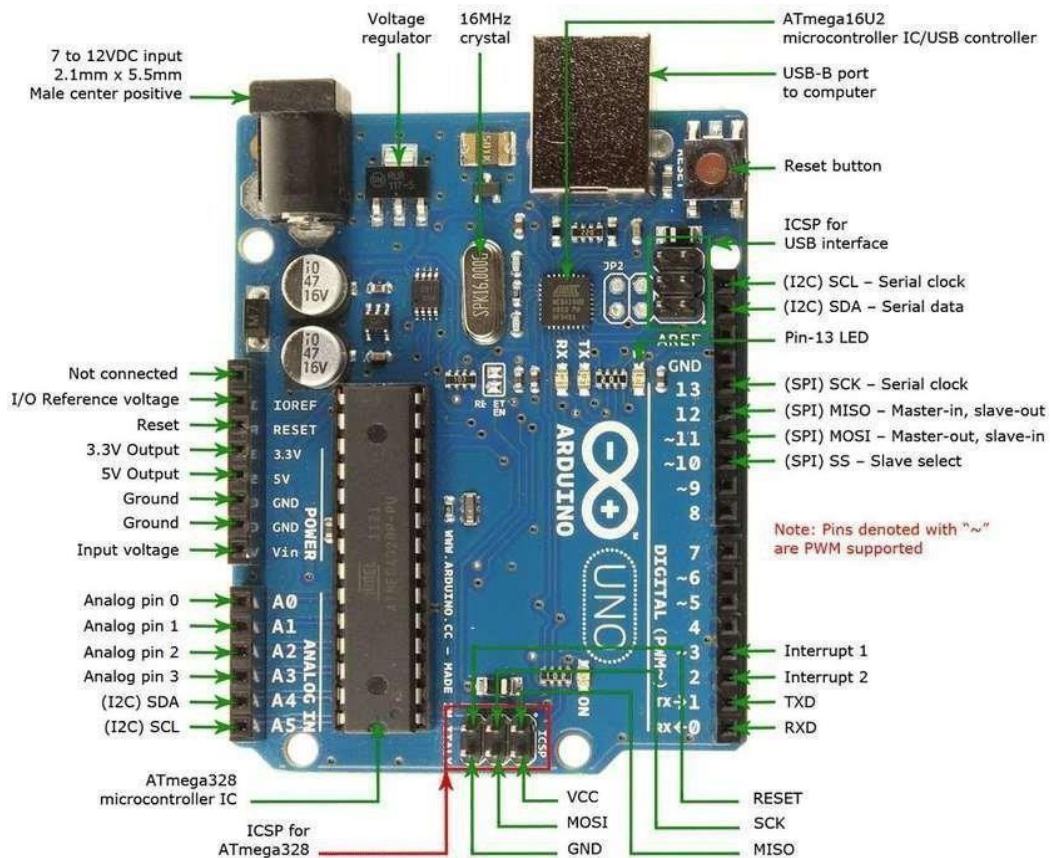


Figure 3.10 Arduino UNO

In the transmitter side of our project, the Arduino Uno is used to process gesture inputs from sensors and transmit control signals wirelessly to the robotic arm. It connects with flex sensors, a gyroscope, and an RF module for seamless communication. The Arduino ensures real-time data processing and reliable transmission for accurate robotic arm movements.

3.8.1 Power Pins

The Arduino Uno can be powered via the USB connection or with an external power supply. The power source is selected automatically.

External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or battery. The adapter can be connected by plugging a 2.1mm center- positive plug into the board's power jack. Leads from a battery can be inserted in the Gnd and Vin pin headers of the POWER connector. The board can operate on an external supply of 6 to 20 volts. If supplied with less than 7V, however, the 5V pin may supply less than five volts and the board may be unstable. If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

The power pins are as follows:

- **VIN.** The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.
- **5V.** This pin outputs a regulated 5V from the regulator on the board. The board can be supplied with power either from the DC power jack (7 - 12V), the USB connector (5V), or the VIN pin of the board (7-12V). Supplying voltage via the 5V or 3.3V pins bypasses the regulator, and can damage your board. We don't advise it.
- **3V3.** A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- **GND.** Ground pins.

3.8.2 Other Pins

- **AREF:** Reference voltage for the analog inputs. Used with analog Reference ().
- **Reset:** (Diecimila-only) Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields that block the one on the board.

3.8.3 Memory

The ATmega328 micro controller features 32 KB of flash memory, with 0.5 KB allocated for the boot loader. Additionally, it includes 2 KB of SRAM for temporary data storage and 1 KB of EEPROM, which allows non-volatile data storage. The EEPROM can be accessed using the EEPROM library, enabling users to read and write data efficiently. These memory specifications make the ATmega328 a reliable choice for embedded systems, ensuring efficient program execution and data retention for various applications.

3.8.4 Input and Output

Each of the 14 digital pins on the Uno can be used as an input or output, using pin Mode (), digital Write (), and digital Read() functions. They operate at 5 volts. Each pin can provide or receive a maximum of 40 mA and has an internal pull-up resistor (disconnected by default) of 20-50 kOhms. In addition, some pins have specialized functions:

- **Serial: 0 (RX) and 1 (TX).** Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip.
- **External Interrupts: 2 and 3.** These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the attach Interrupt () function for details.
- **PWM: 3, 5, 6, 9, 10, and 11.** Provide 8-bit PWM output with the analog Write () function.

3.8.5 Arduino NodeMCU

Arduino NodeMCU is an open-source ESP8266-based development board widely used for IoT and wireless communication projects. It features built-in Wi-Fi connectivity, making it ideal for remote monitoring and automation applications. The board is programmed using the Arduino IDE, supporting both Lua scripting and C++. With GPIO pins, ADC, PWM, and serial communication interfaces, it enables seamless sensor integration and peripheral control. NodeMCU operates on 3.3V, making it compatible with low-power devices. Its compact size, cost-effectiveness, and wireless capabilities make it a preferred choice for smart home automation and industrial IoT projects. In our project, NodeMCU is used on the receiver side to process and execute commands received from the transmitter. It plays a crucial role in wireless communication, receiving gesture data via RF signals and controlling the robotic arm accordingly. With built-in Wi-Fi and GPIO support, it efficiently processes sensor inputs and drives actuators in the NodeMcu's integration.

The NodeMCU's integration with Arduino IDE allows easy programming and customization. Its low power consumption and compact design make it ideal for real-time applications. The pin configuration and connections for the receiver setup are illustrated in Figure 3.11, demonstrating its role in our system.

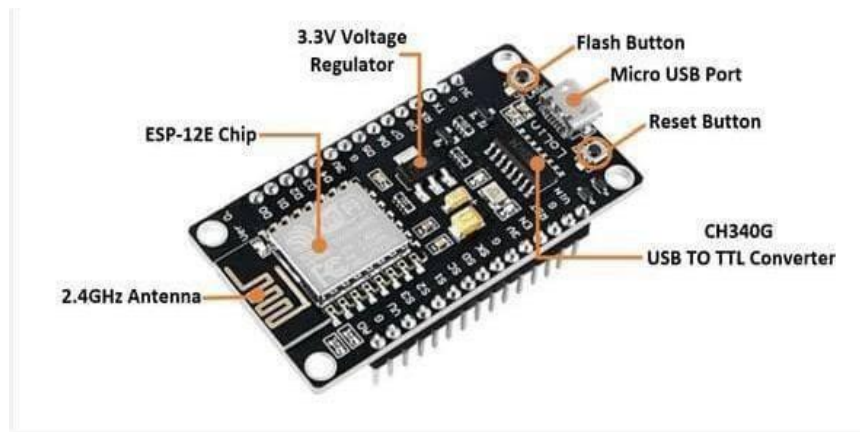


Figure 3.11 Arduino NodeMCU ESP8266

3.8.6 Communication

The Arduino Uno has a number of facilities for communicating with a computer, another Arduino, or other microcontrollers. The ATmega328 provides UART TTL (5V) serial communication, which is available on digital pins 0 (RX) and 1 (TX). An ATmega16U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer.

The '16U2 firmware uses the standard USB COM drivers, and no external driver is needed. However, on Windows, a .inf file is required. The Arduino software includes a serial monitor which allows simple textual data to be sent to and from the Arduino board. The RX and TX LEDs on the board will flash when data is being transmitted via the USB-to-serial chip and USB connection to the computer (but not for serial communication on pins 0 and 1).

A Software Serial library allows for serial communication on any of the Uno's digital pins. The ATmega328 also supports I2C (TWI) and SPI communication. The Arduino software includes a Wire library to simplify use of the I2C bus; see the documentation for details. For SPI communication, use the SPI library.

3.9 LIQUID CRYSTAL DISPLAY (LCD)

A Liquid Crystal Display (LCD) is an electronic display module commonly used in embedded systems for visual output. It operates by manipulating liquid crystals to modulate light, enabling the display of alphanumeric characters and simple graphics. In our project, the LCD is used to display system status, sensor readings, and user inputs, enhancing user interaction. It requires minimal power and provides clear visibility, making it an ideal choice for real-time monitoring. The 16x2 LCD used in our system consists of 16 columns and 2 rows, allowing the display of up to 32 characters at a time. In our project, the Liquid Crystal Display (LCD) is used to visually represent key system operations. It displays gyro sensor movements, indicating left and right motions, and flex sensor commands, showing gripper open and close actions.

Additionally, the fingerprint sensor and keypad sensor are used for authentication, and the entered password is displayed on the LCD for verification. This enhances user interaction by providing real-time feedback on sensor inputs and security operations. The LCD ensures clear monitoring of all essential commands, making the system more user-friendly and efficient. Figure 3.12 shows the LCD display used in the system, illustrating how different sensor inputs and authentication processes are presented to the user.

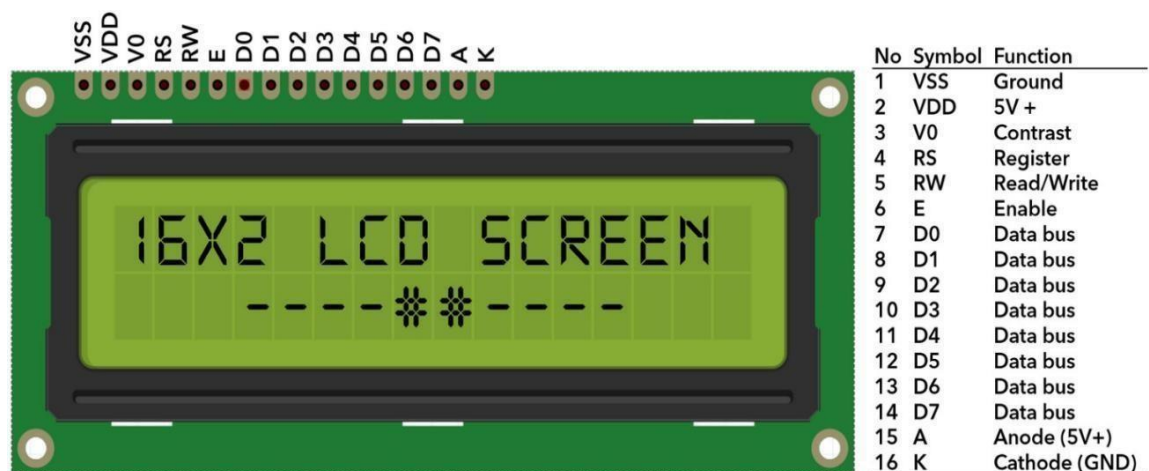


Figure 3.12 16*2 LCD Display

It is one kind of electronic display module used in an extensive range of applications like various circuits & devices like mobile phones, calculators, computers, TV sets, etc. These displays are mainly preferred for multi-segment light-emitting diodes and seven segments. The main benefits of using this module are inexpensive; simple programmable, has animations, and there are no limitations for displaying custom characters, special and even animations, etc.

3.9.1 Pin Configuration

- Pin 1 (Ground/Source Pin): This is a GND pin of display, used to connect the GND terminal of the micro controller unit or power source.
- Pin 2 (VCC/Source Pin): This is the voltage supply pin of the display, used to connect the supply pin of the power source.
- Pin 3 (V0/VEE/Control Pin): This pin regulates the difference of the display, used to connect a changeable POT that can supply 0 to 5V.
- Pin 4 (Register Select/Control Pin): This pin toggles among command or data register, used to connect a microcontroller unit pin and obtains either 0 or 1 (0 = data mode, and 1 = command mode).
- Pin 5 (Read/Write/Control Pin): This pin toggles the display among the read or write operation, and it is connected to a micro controller unit pin to get either 0 or 1 (0 = Write Operation and 1 = Read Operation).
- Pin 6 (Enable/Control Pin): This pin should be held high to execute the Read/Write process, and it is connected to the micro controller unit & constantly held high.
- Pins 7-14 (Data Pins): These pins are used to send data to the display. These pins are connected in two-wire modes like 4-wire mode and 8-wire mode. In the 4-wire mode, only four pins are connected to the micro controller unit like 0 to 3, whereas in the 8-wire mode, 8-pins are connected to the micro controller unit like 0 to 7.

- Pin15 (+ve pin of the LED): This pin is connected to +5V.
- Pin 16 (-ve pin of the LED): This pin is connected to GND.

3.9.2 Registers of LCD

A 16×2 LCD has two registers like data register and command register. The RS (register select) is mainly used to change from one register to another. When the register set is '0', then it is known as command register. Similarly, when the register set is '1', then it is known as data register.

3.9.3 Features of LCD

- The operating voltage of this LCD is 4.7V-5.3V.
- It includes two rows where each row can produce 16 characters.
- The utilization of current is 1mA with no backlight.
- Every character can be built with a 5×8-pixel box.
- The alphanumeric LCDs alphabets & numbers.
- Its display can work on two modes like 4-bit & 8-bit.
- These are obtainable in Blue & Green Backlight.
- It displays a few custom-generated characters.

3.9.4 Command Register

In LCD (Liquid Crystal Display) modules, a command register is a special register used to send commands to control the display's behavior and settings. These commands are sent serially to the LCD module to perform various functions such as clearing the display, setting cursor position, controlling display configuration, and more. Each command corresponds to a specific operation or configuration change.

The command register typically accepts binary values (0s and 1s) or hexadecimal values to specify the command being sent. These commands are predefined by the manufacturer and documented in the LCD data sheet or user manual.

3.9.5 Data Register

The main function of the data register is to store the information that is to be exhibited on the LCD screen. Here, the ASCII value of the character is the information that is to be exhibited on the screen of LCD. Whenever we send the information to LCD, it transmits to the data register, and then the process starts there. When register set =1, then the data register will be selected.

3.10 RADIO FREQUENCY (TX / RX)

Radio Frequency (RF) communication is a wireless transmission technology used in our project to enable seamless data exchange between the transmitter (gesture control module) and the receiver (robotic arm control unit). RF modules operate within specific frequency ranges, commonly 433 MHz or 2.4 GHz, ensuring reliable and long-range communication.

In our project, RF communication plays a crucial role in wirelessly transmitting gesture commands from the wearable controller to the robotic arm. When a user moves their hand, the gyroscope and flex sensors detect motion and convert it into electrical signals. These signals are transmitted using an RF transmitter, and the RF receiver at the robotic arm side decodes them to execute the corresponding movements.

RF technology ensures low latency, high-speed data transfer, and minimal interference, making it an ideal choice for real-time gesture-controlled applications. The TX is an ASK transmitter module. The TX is designed specifically for remote-control, wireless mouse and car alarm system operating at 315/433.92 MHz. The RX is a miniature receiver module that receives On-off keyed (OOK) modulation signal and demodulated to digital signal for the next decoder stage. The RX is designed specifically for remote-control and wireless security receiver operating at 315/434Mhz. UART TTL o/p - Baud Rate – 9600. Figure 3.13 illustrates the RF Communication System used in our project.

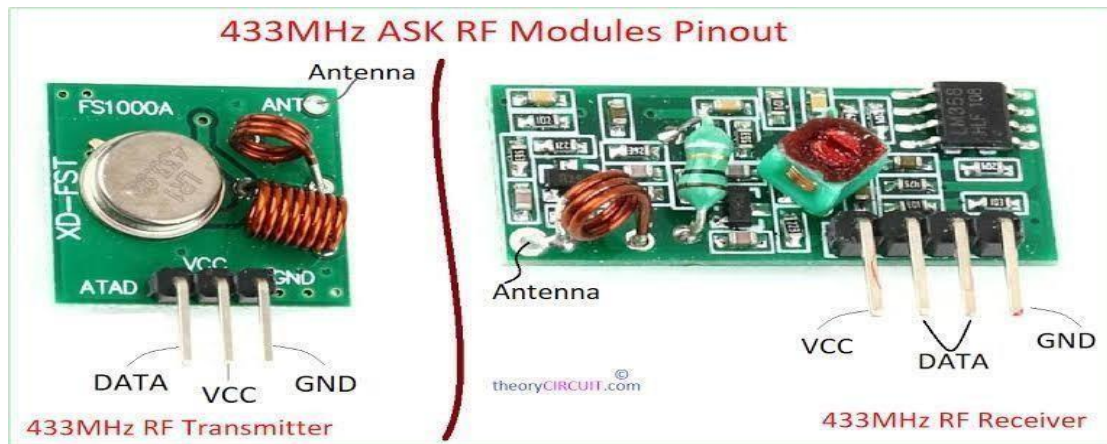


Figure 3.13 Radio Frequencies (TX / RF)

3.10.1 Radio Frequency- Transmitter Side (TX)

In our project, the RF transmitter plays a crucial role in wirelessly sending gesture commands from the user's wearable device to the robotic arm. The transmitter module is integrated with Arduino, which processes signals from the gyroscope and flex sensor. When the user moves their hand, the sensors detect the motion and convert it into an electrical signal, which is transmitted via RF communication.

The RF transmitter ensures low-latency data transfer and operates efficiently over a secure communication channel, preventing signal interference. This wireless setup eliminates the need for physical connections, enabling smooth and precise control of the robotic arm from a distance.

3.10.2 Radio Frequency- Receiver Side (RX)

The Radio Frequency (RF) receiver is crucial for establishing a wireless communication link between the gesture-controlled transmitter and the robotic arm. The receiver is placed on the robotic arm's control unit to decode the signals transmitted via RF communication. It ensures real-time processing of hand gestures captured by sensors like the gyroscope and flex sensor, allowing the robotic arm to perform precise movements. Additionally, the RF receiver works in synchronization with NodeMCU, which processes the received signals and executes the corresponding motor functions.

This setup provides a stable and efficient control mechanism, minimizing delays and maximizing responsiveness in hazardous environments.

3.10.3 Features

- **Wireless Control:** Enables remote operation without physical connections.
- **Long-Range Communication:** Supports distances up to several meters, depending on the RF module used.
- **Real-Time Signal Transmission:** Ensures minimal delay in data transfer for responsive control.
- **Interference Resistance:** Works on dedicated frequency bands to minimize signal disruptions.
- **Low Power Consumption:** Efficient energy usage for battery-powered applications.
- **Reliable Data Transmission:** Uses encoding and decoding techniques to reduce errors.
- **Compact and Cost-Effective:** Small, lightweight, and affordable for various automation applications.
- **Secure Communication:** Can be integrated with encryption for enhanced security.

3.11 POWER SUPPLY

A reliable power supply is a crucial part of any electronic system, ensuring stable voltage and current for efficient operation. In our project, a regulated 5V DC power supply is used to power key components, including the Arduino, NodeMCU, sensors, LCD display, and RF modules. This ensures consistent performance and prevents malfunctions due to voltage fluctuations.

The power supply circuit consists of several essential components: a step-down transformer, bridge rectifier, voltage regulator (7805), and filter capacitors. The step-down transformer converts high-voltage AC into a lower AC voltage, which is then rectified by the bridge rectifier to produce pulsating DC. To smooth out the output, filter capacitors remove voltage fluctuations.

The 7805 voltage regulator ensures a stable 5V DC output, protecting components from potential damage due to voltage variations. For uninterrupted operation, especially in wireless applications, the system supports battery backup. This ensures continuous functionality even during power failures. Rechargeable lithium-ion or lead-acid batteries can serve as backup power sources. Maintaining efficiency in the power supply is crucial, as voltage drops or surges can impact the robotic arm's movement and sensor accuracy.

To enhance reliability, the power supply is carefully designed to prevent overheating and optimize energy efficiency. Heat sinks are used for voltage regulators, ensuring stable performance under continuous operation. A well-regulated power source ensures smooth functioning, making the system ideal for hazardous industrial environments where precision and efficiency are critical.

By providing a stable and uninterrupted power supply, our system guarantees optimal performance, reducing the risk of errors, delays, or failures in industrial automation. This robust power management strategy ensures that the robotic arm operates seamlessly, even in high-risk environments.

3.12 SERVOMOTOR

In our project, servo motors are essential for controlling the precise movements of the wireless robotic arm. These motors are responsible for gripper operation (opening and closing), arm rotation, and positioning based on the hand gestures detected by sensors. The servo motors operate using pulse-width modulation (PWM) signals, allowing for accurate angular control, ensuring smooth and responsive movements.

The robotic arm is controlled using flex sensors and a gyroscope sensor worn by the operator. When the user moves their hand, the flex sensors detect finger bending, while the gyroscope tracks wrist orientation. This data is then processed by the Arduino on the transmitter side and sent via RF communication to the receiver side, where the NodeMCU interprets the signals and drives the servo motors accordingly.

This allows the robotic arm to mimic real-time human hand movements with high precision. Servo motors are used to control the gripper mechanism, ensuring firm yet delicate handling of objects, making the system suitable for applications in hazardous material handling, chemical processing, and industrial automation. The high torque and stability of servo motors enable the robotic arm to perform tasks with precision and reliability, even under challenging conditions.

Additionally, fingerprint authentication and a keypad sensor ensure only authorized users can control the system. The commands given by the user, including gripper movements and arm positioning, are displayed on an LCD screen for real-time feedback.

As shown in Figure 3.14, the servo motors play a critical role in enabling smooth, precise, and secure robotic arm operation, making them indispensable in our project for gesture-controlled automation in hazardous environments.

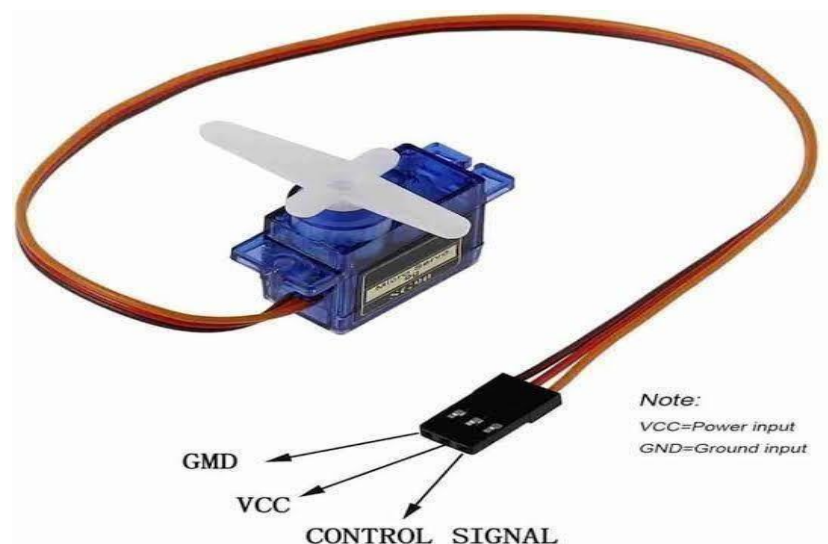


Figure 3.14 Servo Motor

The HandMatic: Gesture-Controlled Wireless Robotic Arm operates based on gesture recognition and wireless communication. The system consists of a transmitter side (worn by the user) and a receiver side (connected to the robotic arm). The flex sensors and gyroscope sensor detect hand and finger movements, converting them into electrical signals. These signals are processed by the Arduino on the transmitter side and transmitted via RF communication to the receiver side.

At the receiver side, the NodeMCU interprets the received signals and controls the servo motors to replicate the user's hand gestures. The flex sensor determines gripper movement (open/close), while the gyroscope controls wrist orientation (left/right movement). For security, fingerprint authentication and a keypad sensor are used to restrict unauthorized access.

The LCD screen provides real-time feedback by displaying executed commands. This system ensures precise, wireless, and secure robotic arm control, making it ideal for hazardous environments.

3.13 GRIPPER MECHANISM

The gripper mechanism in the HandMatic: Gesture-Controlled Wireless Robotic Arm is designed to securely grasp and manipulate objects based on hand gestures, ensuring precise control for handling various materials, especially in hazardous environments.

The flex sensor on the user's glove detects finger movements, converting the bending motion into an electrical signal. This signal is processed by the Arduino on the transmitter side, which then sends the corresponding command to the receiver side via RF communication. Upon receiving the signal, the NodeMCU at the receiver end interprets it and activates the servo motor connected to the gripper mechanism in Figure 3.15.

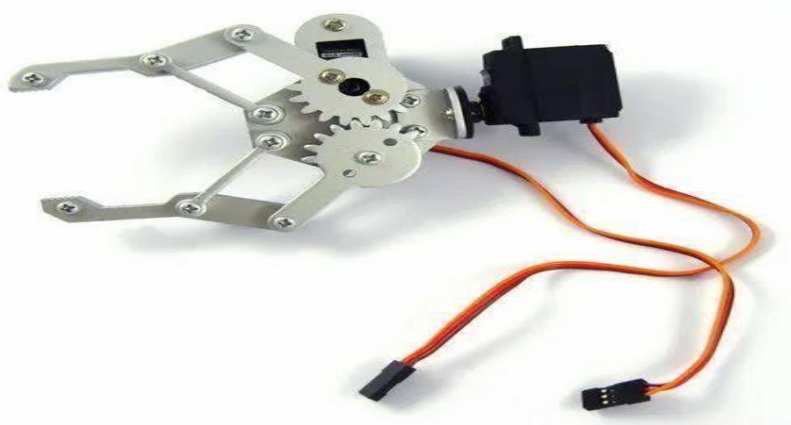


Figure 3.15 Gripper Mechanism

When the user's fingers bend, the servo motor moves the gripper to close, allowing it to firmly grasp an object. Conversely, when the fingers straighten, the servo motor opens the gripper, releasing the object. This enables smooth and intuitive operation, replicating natural hand movements. The gripper mechanism ensures precision, flexibility, and secure handling, making it highly effective for industrial automation, hazardous material handling, and robotic applications.

The HandMatic: Gesture-Controlled Wireless Robotic Arm operates using gesture recognition and wireless communication, ensuring seamless and intuitive control. The system consists of two primary sections: the transmitter side (worn by the user) and the receiver side (connected to the robotic arm).

At the transmitter side, flex sensors and a gyroscope sensor detect hand and finger movements, converting them into electrical signals. These signals are processed by an Arduino microcontroller, which then transmits the data wirelessly using RF communication. The flex sensor determines the gripper's open and close movements, while the gyroscope detects wrist orientation (left/right movement).

At the receiver side, the NodeMCU decodes the received signals and controls servo motors, which move the robotic arm accordingly. To enhance security, fingerprint authentication and a keypad sensor are integrated, ensuring only authorized users can operate the system. An LCD display provides real-time feedback by displaying executed commands. This wireless robotic arm enables precise, secure, and real-time control, making it ideal for industrial automation, hazardous material handling, and remote operations.

3.14 3-AXIS OF ROBOTIC ARM

A 3-axis robot is a robotic system capable of movement along three perpendicular axes-X, Y, and Z-enabling precise control over positioning and manipulation. These robots are commonly used in industrial automation, material handling, robotic assembly, and pick-and-place applications, where efficiency and accuracy are essential. In our HandMatic: Gesture-Controlled Wireless Robotic Arm, the 3-axis movement is achieved through servo motors controlled by gesture recognition.

The gyroscope sensor detects wrist rotations, allowing left and right movement, while the flex sensor determines the gripper's open and close actions. The entire system operates via RF communication, transmitting signals from the transmitter side (wearable device) to the receiver side (robotic arm) for real-time, wireless control.

This system enhances operational flexibility and human-like dexterity, making it ideal for applications in hazardous environments, remote operations, and industrial automation where human presence is risky. The 3-axis design ensures smooth and precise movement, allowing for complex tasks such as object handling, sorting, and assembly. To enhance security, the system integrates fingerprint authentication and a keypad sensor, ensuring only authorized users can control the robotic arm. Additionally, an LCD display provides real-time feedback, showing executed commands and system status for improved monitoring.

The 3-axis mechanism in the HandMatic robotic arm significantly improves precision, efficiency, and control, making it a reliable solution for automated robotic applications. The 3-axis movement of the robotic arm is illustrated in Figure 3.16, highlighting its capability in real-world operations.

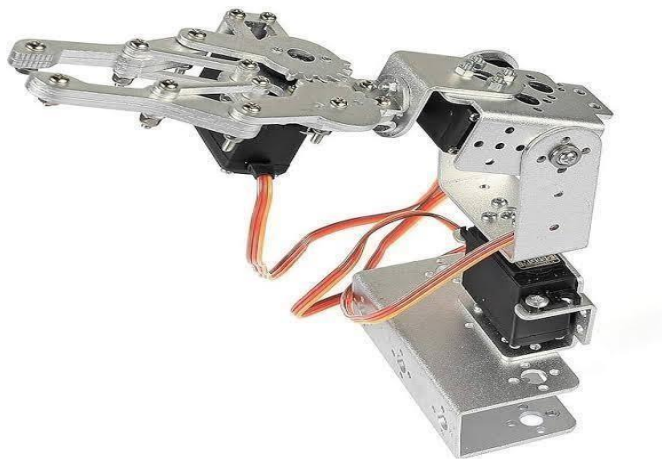


Figure 3.16 3- Axis Robotic Arm

The HandMatic: Gesture-Controlled Wireless Robotic Arm operates based on gesture recognition, wireless communication, and secure authentication. The system consists of two primary sections: the transmitter side (wearable device) and the receiver side (robotic arm).

On the transmitter side, flex sensors and a gyroscope sensor capture the user's hand and wrist movements. The flex sensors detect finger bending to control the gripper (open/close), while the gyroscope measures wrist orientation to control left and right movements. These sensor readings are processed by the Arduino, which converts them into electrical signals. The data is then transmitted wirelessly via RF communication to the receiver side.

On the receiver side, a NodeMCU micro controller processes the received signals and controls servo motors to replicate the user's hand movements in real time. The system also integrates fingerprint authentication and a keypad sensor to ensure only authorized users can operate the robotic arm.

An LCD display provides real-time feedback by displaying executed commands, enhancing user interaction and monitoring. This ensures precise and responsive control, making the robotic arm ideal for hazardous environments and industrial automation. The combination of gesture-based control, wireless transmission, and security features makes the HandMatic robotic arm an efficient, user-friendly, and safe solution for remote robotic operations.

3.15 BATTERY

In our HandMatic: Gesture-Controlled Wireless Robotic Arm project, different types of batteries are used on the transmitter and receiver sides to ensure reliable power management and smooth operation.

3.15.1 Transmitter Side Battery

The transmitter side, which includes the Arduino, flex sensors, gyroscope, and RF transmitter module, requires a compact and lightweight power source. A Li-ion (Lithium-Ion) battery or Li-Po (Lithium-Polymer) battery is used, typically with a voltage rating of 3.7V to 7.4V. These batteries offer high energy density, recharge ability, and longer operation time, making them suitable for wearable applications. Figure 3.17, Since the transmitter system primarily processes hand gestures and transmits signals, the power consumption is relatively low, and a small-capacity battery is sufficient for extended usage.



Figure 3.17 Transmitter Side Battery

3.15.2 Receiver Side Battery

The receiver side, which includes the NodeMCU, servo motors, LCD, fingerprint sensor, and keypad, requires a more robust power source due to the higher energy demands of motors and processing components. A 12V DC rechargeable battery is used, with a sufficient current rating of 2A to 5A to support continuous operation. Figure 3.18, Servo motors, especially those controlling the robotic arm movements and gripper mechanism; require stable power to ensure smooth and precise motion.



Figure 3.18 Receiver Side Battery

3.16 VOLTAGE REGULATOR INTEGRATED CIRCUIT (IC)

In this project, a voltage regulator integrated circuit (IC) is used on both the transmitter and receiver sides to ensure a stable power supply for all components, as shown in Figure 3.19. Voltage regulators play a crucial role in maintaining a constant output voltage, protecting sensitive components from voltage fluctuations, and preventing damage due to voltage spikes or drops.

On the transmitter side, a 7805 voltage regulator is used to step down and stabilize the input voltage to 5V, powering the Arduino, gyro sensor, flex sensors, and RF transmitter module. The 7805 ensures consistent voltage, preventing fluctuations that could affect data transmission and sensor accuracy.

On the receiver side, another 7805 voltage regulator provides a steady 5V output to the NodeMCU, RF receiver module, fingerprint sensor, and motor driver circuit. The motor driver (L298N) requires a regulated supply to efficiently operate the servomotors. Since motors can cause voltage variations due to load changes, the voltage regulator stabilizes the system's performance.

By integrating voltage regulator ICs on both sides, the project ensures reliable operation, prevents overheating, and enhances efficiency in the wireless robotic arm system.

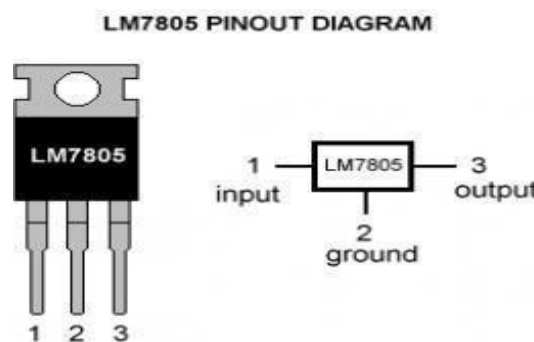


Figure 3.19 Voltage Regulator Integrated Circuit (7805 5V)

3.17 SUMMARY

This chapter introduces HandMatic: Gesture-Controlled Wireless Robotic Arm, designed for hazardous environments like chemical processing and toxic material handling. Using RF communication, it enables remote operation, enhancing safety and efficiency. The limitations of existing systems, such as gesture inaccuracy, user fatigue, limited control inputs, complex training, and restricted task execution, highlight the need for an improved solution. HandMatic addresses these challenges by integrating gyroscopes and flex sensors for precise gesture recognition and incorporating fingerprint authentication and keypad security to restrict access to authorized users. The transmitter side captures hand movements and sends commands via RF, while the receiver side processes these signals to control the robotic arm with high accuracy. Key components, including gyroscope and flex sensors, provide human-like dexterity, making HandMatic a secure, efficient, and intuitive solution for industrial automation in hazardous environments.

CHAPTER 4

HARDWARE IMPLEMENTATION AND RESULTS

4.1 INTRODUCTION

The hardware implementation of this project involves assembling and integrating various components to create a functional gesture-controlled wireless robotic arm. The system consists of a transmitter and receiver unit, incorporating flex sensors, a gyroscope, an Arduino, NodeMCU, servo motors, an RF module, a fingerprint sensor, and a keypad for security. The implementation ensures real-time, accurate movement replication based on hand gestures. The results demonstrate precise control, enhancing safety and efficiency in hazardous environments while ensuring secure operation through authentication mechanisms.

4.2 CIRCUIT DIAGRAM OF PROPOSED SYSTEM

In this system, an Arduino acts as the central controller, receiving input from a gyro sensor, fingerprint sensor, and keypad. The gyro sensor is connected via I2C to detect orientation or movement, while the fingerprint sensor communicates through UART to authenticate users. The keypad allows for manual input of commands. The Arduino processes these inputs and wirelessly sends commands to a remote system via an RF transmitter. On the receiving end, an RF receiver transmits the data to a motor driver, which controls the motor's operation based on the received signals, such as direction and speed.

Power for the entire system is provided by a battery, ensuring the setup is portable and functional without requiring external power sources. The motor driver, such as the L298N, is used to interface between the Arduino and the motor, enabling precise control of the motor's actions based on the processed data from the sensors, as shown in Figure 4.1.

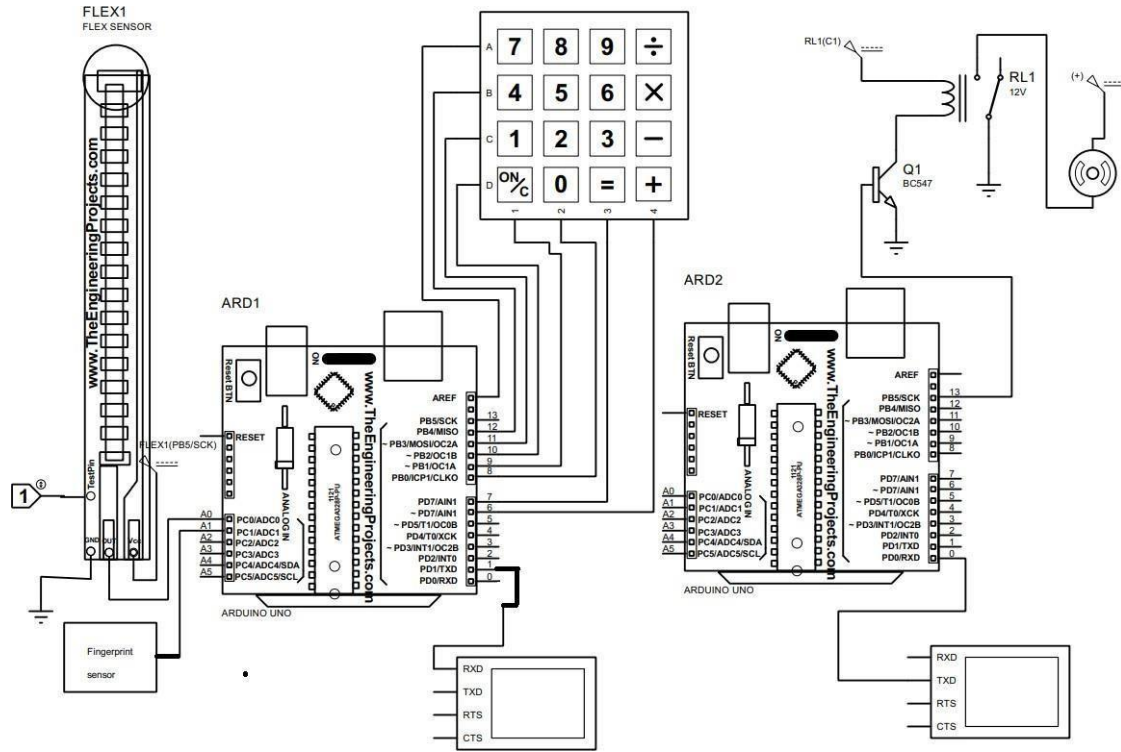


Figure 4.1 Circuit Diagram of Proposed system

4.3 EXPERIMENTAL SETUP AND RESULTS

The experimental setup of the HandMatic: Gesture-Controlled Wireless Robotic Arm consists of a transmitter and receiver system for seamless wireless communication. The transmitter side includes an Arduino interfaced with a gyroscope and flex sensors to capture hand movements, converting them into electrical signals. These signals are transmitted wirelessly via an RF module to the receiver unit. The receiver side, powered by NodeMCU, processes the incoming data and controls servo motors to execute the corresponding robotic arm movements. For security, the system integrates a fingerprint sensor and a keypad, ensuring only authorized users can operate the robotic arm. An LCD screen provides real-time feedback by displaying executed commands. The results show smooth and accurate robotic arm control with minimal response delay, confirming its effectiveness in handling hazardous tasks. The system ensures precision, safety, and reliability, making it suitable for industrial applications.

In our system Figure 4.2, Arduino processes inputs from a gyro sensor, fingerprint sensor, and keypad to control a motor via RF communication. It transmits commands wirelessly to a receiver, ensuring secure and flexible operation. A battery powers the system, enabling portability and standalone functionality in hazardous environments.

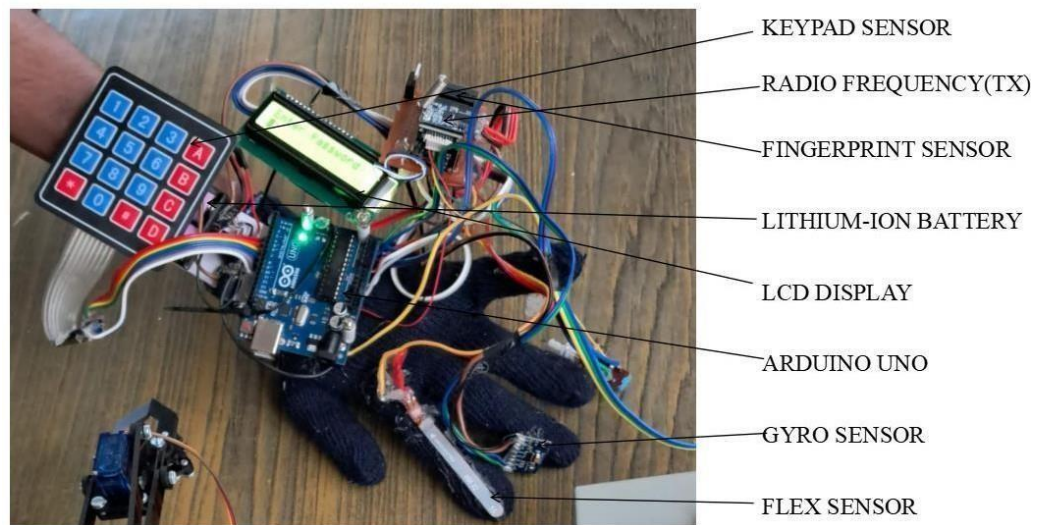


Figure 4.2 Transmitter Side of Proposed Side

The RF receiver collects data and sends it to a motor driver, which controls the motor's start/stop, speed, and direction. Figure 4.3, This system supports robotics, remote control, and security applications. Battery power ensures portability and independence from fixed sources, making it ideal for mobile and outdoor use.

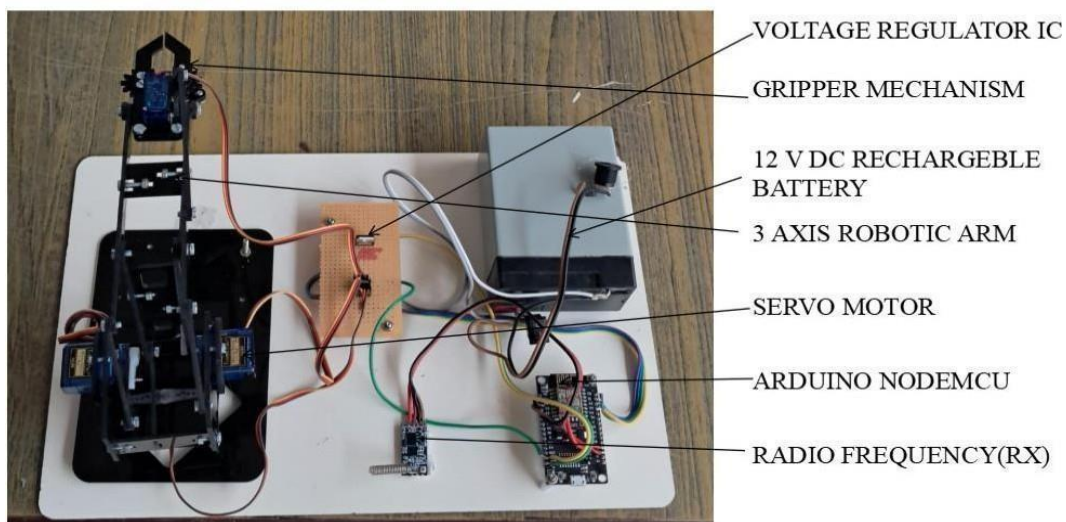


Figure 4.3 Receiver Side of Proposed System

The keypad sensor is used for secure access by requiring users to enter the password "1234#" (Figure 4.4). If the correct password is entered, access is granted; otherwise, it is denied. The LCD display shows real-time feedback, ensuring security, reliability, and controlled access to the robotic arm system.



Figure 4.4 Enter Password in Keypad Sensor

In our project, the fingerprint sensor is used for secure authentication by requiring the user to place their finger on the sensor (Figure 4.5). Once the fingerprint is verified, access is granted, allowing the user to wear the gesture-control glove. The LCD display confirms successful authentication by showing an access message (Figure 4.6). After authentication, the user can control the robotic arm through hand gestures, ensuring secure and intuitive operation for industrial and hazardous environments.



Figure 4.5 Place Finger in Fingerprint Sensor



Figure 4.6 Access is Granted to Allowing the Robotic Arm

In our project, the LCD display shows real-time updates when the robotic arm picks and places materials Figure 4.7. When the gesture-controlled robotic arm successfully grasps an object using the gripper mechanism, the LCD displays "Object Picked." After placing the object in the desired location, it updates to "Object Placed." This feedback ensures accurate operation, helping users monitor the arm's actions and improve precision in material handling tasks, especially in hazardous or industrial environments.



Figure 4.7 Pick and Place Mechanism

4.4 RESULT AND DISCUSSION

The results of the HandMatic: Gesture-Controlled Wireless Robotic Arm demonstrates the system's efficiency in remotely controlling a robotic arm through hand gestures. The integration of flex and gyroscope sensors successfully translates hand movements into precise robotic actions, ensuring smooth and accurate operation. The RF communication between the transmitter and receiver ensures seamless data transmission with minimal latency, allowing real-time control. Security features, including fingerprint authentication and keypad access, effectively restrict unauthorized usage, enhancing system safety. The LCD provides real-time feedback by displaying executed commands, improving user interaction. The robotic arm accurately performs gripping, rotation, and movement tasks, proving its suitability for hazardous environments.

The overall system is reliable, secure, and user-friendly, making it applicable in industrial automation, chemical handling, and remote operations where human presence is risky.

4.5 APPLICATIONS

The HandMatic: Gesture-Controlled Wireless Robotic Arm has diverse applications across various industries, enhancing safety, efficiency, and automation.

- **Industrial Automation** – Used in hazardous environments such as chemical plants, nuclear facilities, and manufacturing units to handle dangerous materials remotely.
- **Medical and Rehabilitation** – Assists physically challenged individuals in performing tasks through gesture-based control.
- **Military and Defense** – Applied in bomb disposal, surveillance, and remote- controlled operations in dangerous zones.
- **Space Exploration** – Helps astronaut scontrol robotic arms in space stations or planetary rovers.
- **Agriculture** – Automates precision tasks like crop handling and pesticide spraying without human exposure.
- **Education and Research** – Used for robotics training, experimentation, and technological advancements in automation.
- **Home Automation** – Helps in controlling appliances and performing household tasks using gestures.

4.6 SUMMARY

This chapter focuses on the hardware implementation and results of the HandMatic robotic arm. The system utilizes Arduino to process sensor inputs, including gyroscope and flex sensors, and transmits commands wirelessly via RF modules. On the receiver side, NodeMCU interprets these signals to control servo motors, ensuring smooth robotic arm movement. Security is reinforced through fingerprint authentication and a keypad, preventing unauthorized access. The results demonstrate the system's ability to provide precise, efficient, and reliable robotic arm control, making it ideal for hazardous industrial applications and remote operations.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The HandMatic: Gesture-Controlled Wireless Robotic Arm successfully demonstrates an advanced robotic system capable of performing tasks based on intuitive human gestures. By integrating wireless RF communication, sensor-based control, and multi-layer security, the project offers a safe and efficient solution for hazardous environments where human intervention is risky.

The system's transmitter side utilizes flex sensors and a gyroscope to capture hand and finger movements, converting them into electrical signals. These signals are processed by the Arduino and sent wirelessly to the receiver side using RF communication. At the receiver end, NodeMCU interprets these signals, directing the servo motors to replicate the hand gestures in real time. The LCD display provides instant feedback on executed commands, ensuring precise operation. For security; the robotic arm incorporates fingerprint authentication and a keypad sensor, restricting access to authorized users only. This multi-factor authentication enhances system reliability, making it suitable for industrial automation, medical applications, and remote hazardous operations.

The project effectively demonstrates the integration of robotics, wireless control, and human-computer interaction, offering a highly intuitive and user-friendly robotic control system. The use of voltage regulators ensures stable power distribution across both transmitter and receiver sides, preventing fluctuations and maintaining system efficiency. By eliminating the need for direct human involvement in dangerous environments, the robotic arm reduces risks associated with handling toxic materials, high-temperature zones, or hazardous machinery. It can also be extended to assistive technology, prosthetics, and smart automation applications.

In conclusion, the HandMatic robotic arm provides a scalable and innovative approach to gesture-controlled automation, demonstrating how modern robotics can enhance safety, precision, and efficiency in industrial and real-world applications. Future improvements could include IoT integration, AI-based gesture recognition, and machine learning algorithms for even greater adaptability and performance.

5.2 FUTURE SCOPE

The HandMatic: Gesture-Controlled Wireless Robotic Arm has the potential to evolve with advancements in robotics, artificial intelligence (AI), and IoT (Internet of Things). Future trends in this field will focus on enhancing automation, precision, and adaptability for various industrial and real-world applications.

1. Integration with AI and Machine Learning

- AI-based gesture recognition can improve the accuracy of hand gesture detection, reducing errors in robotic arm movements.
- Machine learning algorithms can enable the system to adapt to different users' gestures, enhancing personalization and efficiency.

2. IoT-Enabled Remote Control

- The system can be upgraded to support IoT connectivity, allowing real-time monitoring and control over the internet.
- Cloud-based storage and analytics can improve data management and performance tracking.

3. Enhanced Wireless Communication

- Moving from RF-based transmission to Wi-Fi, Bluetooth, or 5G can offer greater reliability, lower latency, and extended range.
- Improved communication can enable multi-robot coordination for complex tasks in industrial automation.

4. Advanced Security Features

- Block chain- based authentication can be incorporated for secure access control.
- Biometric authentication, including facial or iris recognition, can be integrated for even higher security.

5. Human-Robot Collaboration

- Future systems could focus on collaborative robotics (Co-bots), where robots and humans work together in industrial and medical applications.
- Haptic feedback technology could allow users to feel resistance or pressure, improving control precision.

6. Medical and Assistive Applications

- Gesture-controlled robotic arms could be used in prosthetics and rehabilitation devices for people with disabilities.
- Remote surgical robotics could be developed for enhanced precision in the medical field.

By incorporating these future trends, the HandMatic robotic arm can become a versatile, intelligent, and globally connected robotic solution, expanding its applications across various domains.

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APPENDICES

CODE FOR FINGER PRINT AND KEYPAD SENSOR

```
#include <Adafruit_Fingerprint.h>
#include <SoftwareSerial.h> #include
<Wire.h> #include
<math.h>

#include <Bonezegei_LCD1602_I2C.h>

#include <Keypad.h>

int z1=0; int
y1=0;

const int ADXL345_ADDR = 0x53; int16_t x,
y, z;

long xSum = 0, ySum = 0, zSum = 0; int
sampleCount = 100;

float xyAngle = 0;

// LCD setup Bonezegei_LCD1602_I2C
lcd(0x27);

// SoftwareSerial for fingerprint sensor communication (pins 2 and 3) SoftwareSerial
mySerial(2, 3);

Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial); const
char* correctPasscode = "1234"; // Predefined 4-digit passcode String
enteredPasscode = ""; // Store user input

// Define the rows and columns of the keypad
```

```

const byte ROW_NUM      = 4; // Four rows

const byte COL_NUM      = 4; // Four columns

char keys[ROW_NUM][COL_NUM] = {

  { '1', '2', '3', 'A' },

  { '4', '5', '6', 'B' },

  { '7', '8', '9', 'C' },

  { '*', '0', '#', 'D' }

};

// Define the row and column pins on the Arduino byte

pin_rows[ROW_NUM] = {11, 10, 9, 8};

byte pin_cols[COL_NUM] = {7, 6, 5, 4};

// Create an instance of the Keypad library

Keypad keypad = Keypad(makeKeymap(keys), pin_rows, pin_cols, ROW_NUM,
COL_NUM);

void setup() {

  // Start serial communication

  Serial.begin(9600); Wire.begin();

  while (!Serial); // For Yun/Leo/Micro/Zero/... delay(100);

  // Serial.println("\n\nAdafruit Fingerprint sensor");

  Wire.beginTransmission(ADXL345_ADDR);

  Wire.write(0x2D);

  Wire.write(0x08); Wire.endTransmission();

  lcd.begin(); lcd.print("FINGER PRINT&

PW");

```

```

lcd.setPosition(0, 1);

lcd.print("BASED ARM"); //param1 = X          param2 = Y

delay(2000);

lcd.setPosition(0, 0);

//Serial.println("ADXL345 Initialized");

delay(500);

// Initialize the fingerprint sensor

finger.begin(57600);

if (finger.verifyPassword()) {

lcd.clear();  lcd.print("Sensor

Found!");

} else {

lcd.clear();

lcd.print("Sensor Error");

while (1); // Stop if the sensor is not found

}

delay(1000); void

loop() {

char key = keypad.getKey();

lcd.clear(); lcd.setPosition(0, 0);

lcd.print("Enter Password");

lcd.setPosition(0, 1);

lcd.print(enteredPasscode.c_str());

delay(100); if

(key) {

```

```

// Serial.print("Key Pressed: ");

//Serial.println(key);

// Handle special keys for input
if (key == '#') { // Submit the entered code checkPasscode();
}

else if (key == '*') { // Clear input (if needed)
enteredPasscode = "";
// Serial.println("Input cleared.");
}

else { // Add pressed key to the input
if (enteredPasscode.length() < 4) { // Only accept 4 digits
enteredPasscode += key;
// Serial.print("Entered: ");
//Serial.println(enteredPasscode);
}
}
}

while(z1)
{
verifyFingerprint();
}

while(y1)
{
lcd.clear();

```

```

lcd.setPosition(0, 0); lcd.print("PICK
AND"); lcd.setPosition(0, 1);
lcd.print("PLACE");          //param1 = X  param2 = Y
delay(100);
calculateAverage();
calculateXYAngle();
compareValues(); flex();
}

delay(100);
}

void verifyFingerprint() { int p
= -1;

//Serial.println("Place Finger to Verify");

// Wait for a valid finger image while (p
!= FINGERPRINT_OK) {
p = finger.getImage(); switch (p)
{
case FINGERPRINT_OK:
lcd.clear(); lcd.print("Image
taken"); break;
case FINGERPRINT_NOFINGER:
lcd.clear(); lcd.print("No
Finger"); break;

```



```

case FINGERPRINT_PACKETRECEIVEERR:

lcd.clear(); lcd.print("Comm

Error"); break;

case FINGERPRINT_IMAGEFAIL:

lcd.clear(); lcd.print("Image

Error"); break;

default:

lcd.clear(); lcd.print("Unknown

Error"); break;

}

delay(100);

}

// Convert image to template and search for a match p =

finger.image2Tz();

if (p == FINGERPRINT_OK) {

p = finger.fingerSearch();

if (p == FINGERPRINT_OK) {

lcd.clear(); lcd.print("Access

Granted"); lcd.setPosition(0, 1);

lcd.print("USE UR GLOVE");

//lcd.print(finger.fingerID); y1=1;

z1=0;

}

else      {

y1=0;

```

```

z1=0;

lcd.clear();

lcd.print("No Match Found");

}

} else {

lcd.clear();

lcd.print("Error Converting");

}

delay(2000); // Wait for 2 seconds before verifying again

}

void checkPasscode() {

// Compare the entered passcode with the correct passcode if

(enteredPasscode == correctPasscode) {

z1=1;

lcd.clear();

lcd.setPosition(0, 0);

lcd.print("PASSWORD OK");

delay(2000);

lcd.clear();

lcd.setPosition(0, 0);

lcd.print("PLACE FINGER ");

lcd.setPosition(0, 1);

lcd.print("ON SENSOR");           //param1 = X  param2 = Y

delay(2000);

```

```

//Serial.println("\nPasscode Correct! Access Granted.")

} else {
z=0;
//Serial.println("\nIncorrect Passcode. Access Denied.");
}

// Reset entered passcode after check enteredPasscode = "";
//Serial.println("Please enter the 4-digit passcode again.");
}

void calculateAverage() { xSum = 0; ySum
= 0;
zSum = 0;

for (int i = 0; i < sampleCount; i++) { readADXL345(); xSum
+= x; ySum += y; zSum += z; delay(10);
}

x = xSum / sampleCount;

y = ySum / sampleCount;

z = zSum / sampleCount;

}

void readADXL345(

{ Wire.beginTransaction(ADXL345_ADDR); Wire.write(0x32);
Wire.endTransmission(false); Wire.requestFrom(ADXL345_ADDR, 6, true); x =
(Wire.read() | (Wire.read() << 8));
y = (Wire.read() | (Wire.read() << 8));
z = (Wire.read() | (Wire.read() << 8));

}

```

```

void calculateXYAngle() {

xyAngle = atan2(y, x) * 180.0 / M_PI; if (xyAngle < 0) {
xyAngle += 360;
}

}

void compareValues() {

if (abs(x) > abs(y) && abs(x) > abs(z)) { if (x > 0) {

//Serial.println("Orientation: Right Tilt"); lcd.setPosition(0, 1); lcd.print("Right Tilt ");
Serial.println("D");

} else {

// Serial.println(

Orientation: Left Tilt"); lcd.setPosition(0, 1); lcd.print("Left Tilt "); Serial.println("C");

}

} else if (abs(y) > abs(x) && abs(y) > abs(z)) { if (y > 0) {

// Serial.println("Orientation: Front Tilt"); lcd.setPosition(0, 1); lcd.print("Front Tilt ");
Serial.println("A");

} else {

// Serial.println("Orientation: Back Tilt"); lcd.setPosition(0, 1); lcd.print("Back Tilt ")

Serial.println("B");

}

} else if (abs(z) > abs(x) && abs(z) > abs(y)) { if (z > 0) {

// Serial.println("Orientation: Normal"); lcd.setPosition(0, 1); lcd.print("Normal ");

} else {

```

```

// Serial.println("Orientation: Upside Down"); lcd.setPosition(0, 1); lcd.print("Upside
Down");

}

} else {

//Serial.println("Orientation: Unknown"); lcd.setPosition(0, 1); lcd.print("Unknown ");

}

}

void flex()

{

float v = analogRead(A0);

// Convert the analog reading (which goes from 0 - 1023) to a voltage (0 - 5V): float v1
= v * (5.0 / 1023.0);

// print out the value you read:

Serial.println(v1);

// Wait for a short period before taking the next reading delay(100);

if(v1>1.2)

{

Serial.println("F");

}

else

{

Serial.println("E")

```

CODE FOR SERVO MOTOR AND GRIPPER

```
#include <Servo.h>

// Declare servo objects for each axis

Servo servo1; // Left-Right rotation (Servo 1)

Servo servo2; // Front-Back arm movement (Servo 2) Servo servo3; // Gripper open-
close (Servo 3)

// Initialize positions for each servo

int pos1 = 90; // Default position for Servo 1 (Left-Right) int pos2 = 90; // Default
position for Servo 2 (Front-Back) int pos3 = 90;
// Default position for Servo 3 (Gripper) void setup() {

Serial.begin(9600); // Start serial communication at 115200 baud rate

// Attach the servos to the corresponding pins on NodeMCU

servo1.attach(D0); // Left-Right rotation servo on pin D2 servo2.attach(D1); // Front-
Back movement servo on pin D3 servo3.attach(D2); // Gripper servo on pin D4

// Initialize servos to default positions servo1.write(pos1); servo2.write(pos2);
servo3.write(pos3);

}

void loop() {

// Check if there is any incoming data from Serial if (Serial.available() > 0) { char
command = Serial.read(); // Read the incoming byte as a command

// Handle the serial commands for different actions if (command == 'A') { // Front
movement

pos2 = min(pos2 + 10, 180); // Increase position to move the arm front (max 180
degrees)

servo2.write(pos2);
```

```

}

else if (command == 'B') { // Back movement

pos2 = max(pos2 - 10, 0);      // Decrease position to move the arm back (min 0
degrees)

servo2.write(pos2);

}

else if (command == 'C') { // Left rotation

pos1 = max(pos1 - 10, 0);      // Rotate left (min 0 degrees) servo1.write(pos1);

}

else if (command == 'D') { // Right rotation

pos1 = min(pos1 + 10, 180); // Rotate right (max 180 degrees) servo1.write(pos1);

}

else if (command == 'E') { // Open gripper

pos3 = 180; // Set servo to open position (fully open) servo3.write(pos3); else

if (command == 'F') { // Close gripper

pos3 = 68; // Set servo to close position (fully closed) servo3.write(pos3);

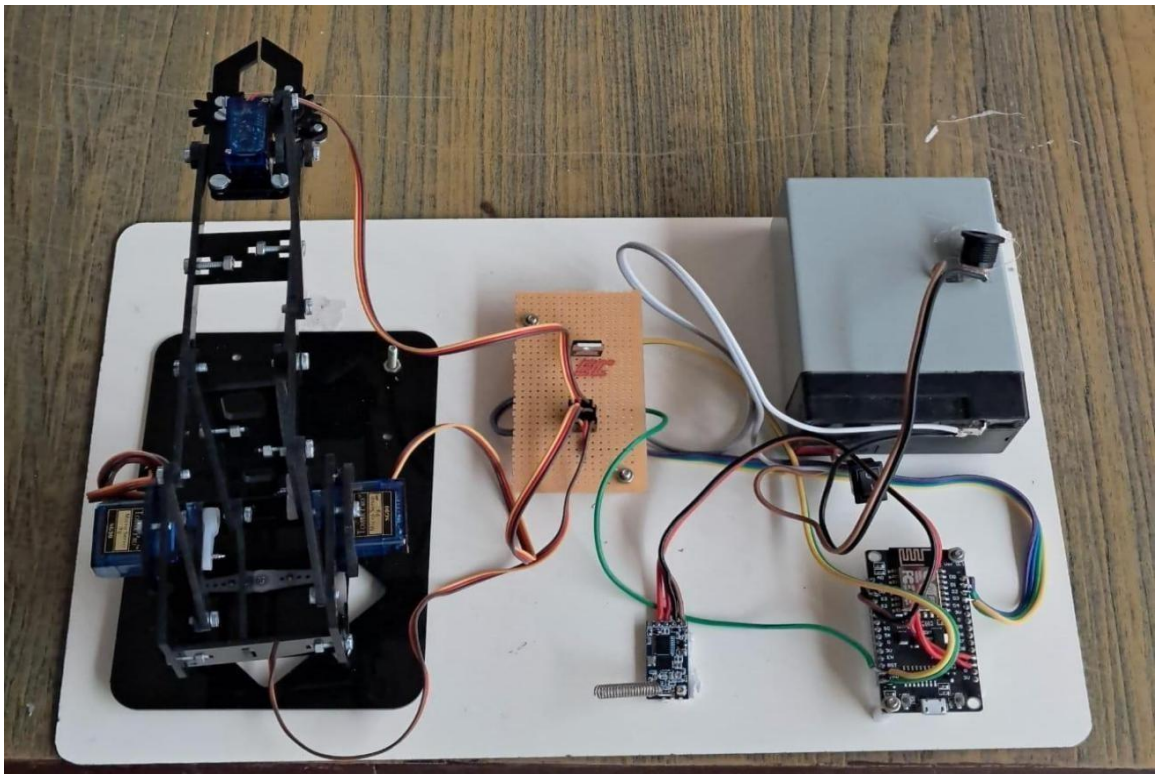
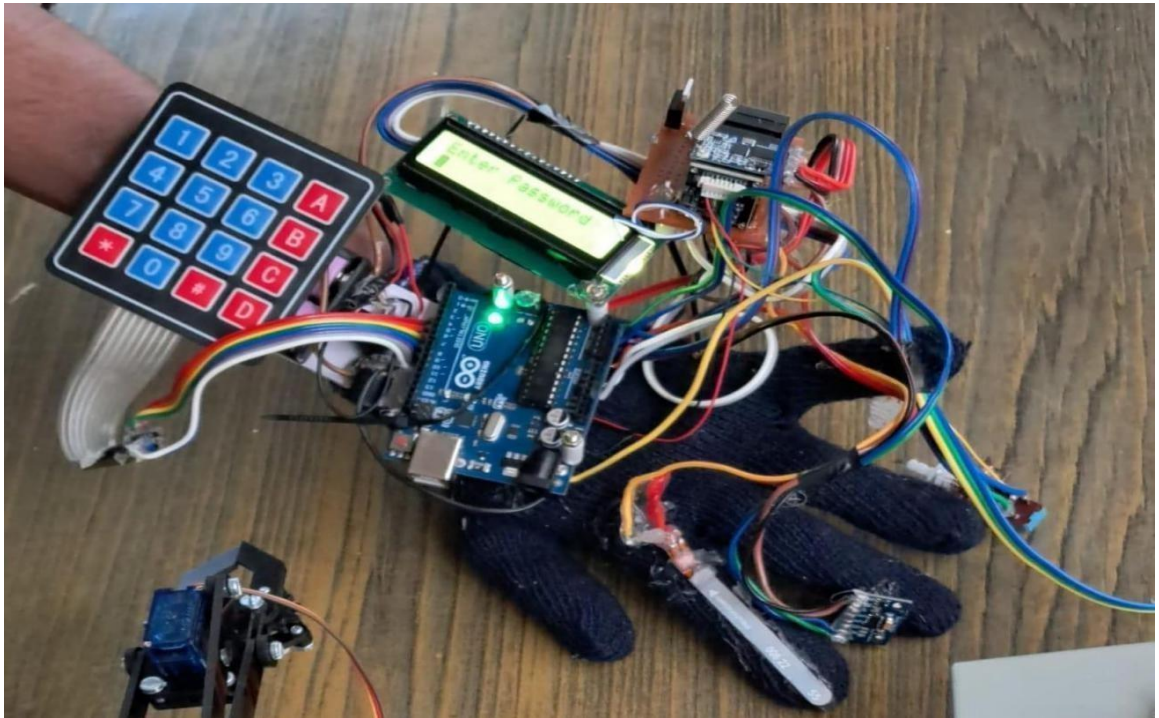
}

delay(50); // Small delay to allow for servo movement

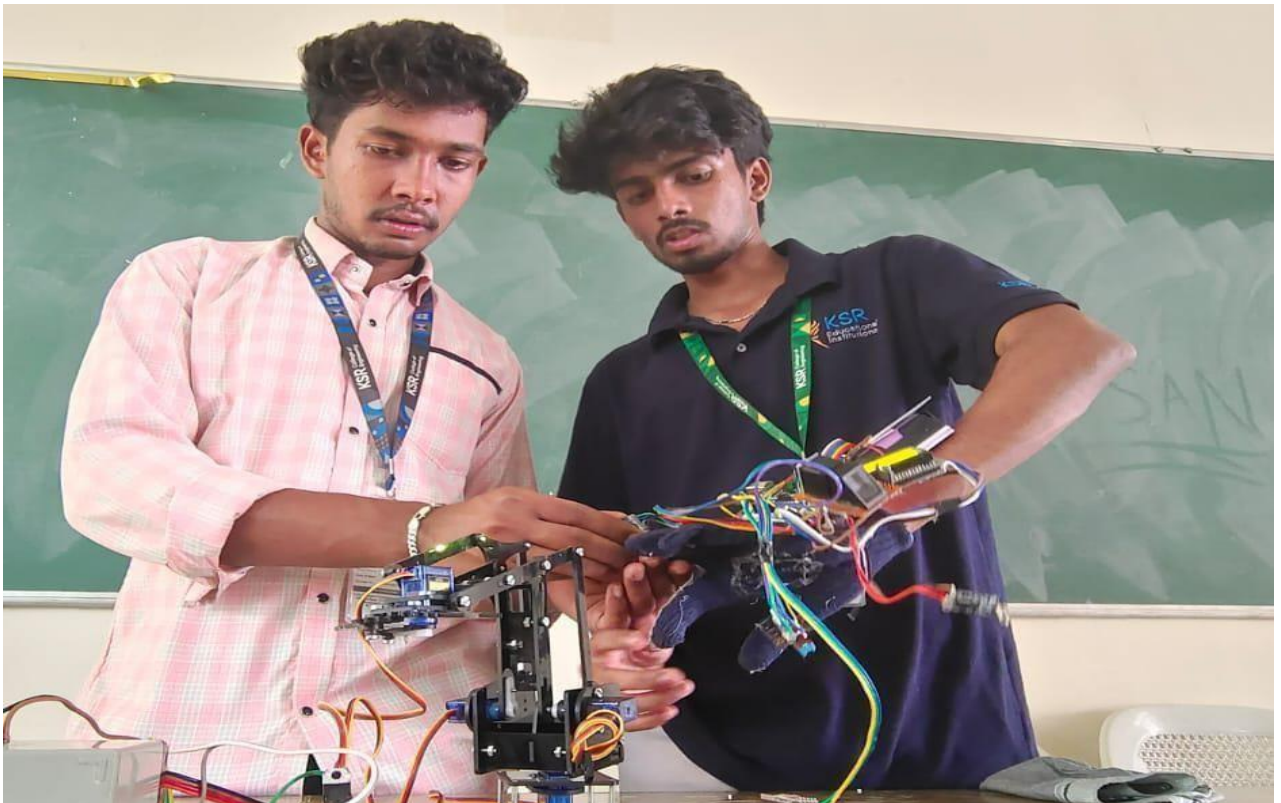
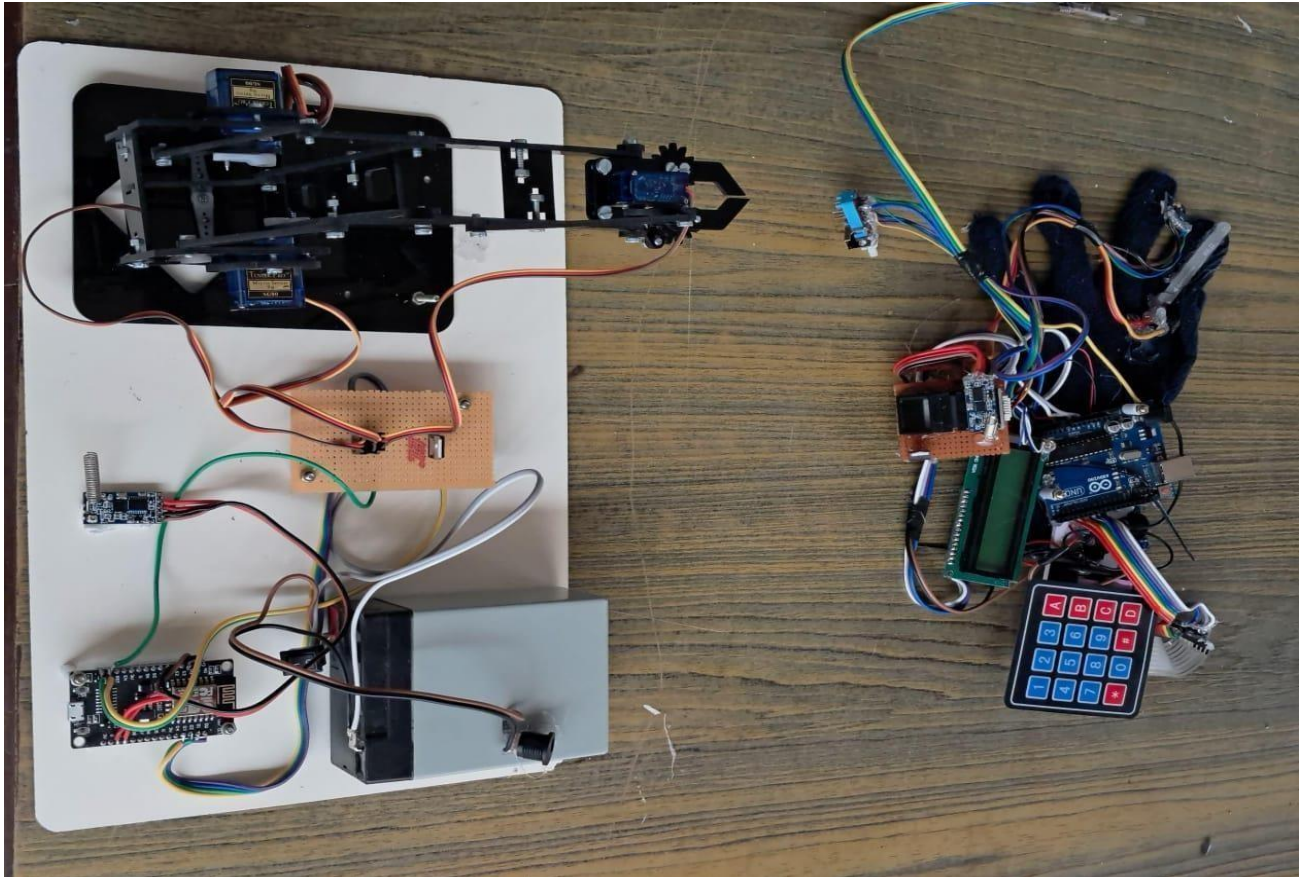
}

```

PHOTOGRAPHS



PHOTOGRAPHS



PUBLICATION

1. **Mr.S.Dhivagar, K.Akash, M.Boopathi, D.Yuvanesh, S.Kaviyarasu,**
“Handmatic: Wireless Robotic Arm Control Through Gesture Recongnition”,
Selected in International Conference on Optics and Communication Technology
(ICOCT-2025), Jythoy Institute of Technology, Bangalore going to happen on 13
June 2025.